



User Interface Design (23CSE113)

LAB MANUAL

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INDEX

S. No	Title	Page No.	REMARKS
1.	Week-1 1A. To understand the installation process of a VS code and get familiar with its environment for web development. 1B. To learn the basic structure of an HTML document and create a simple webpage using headings and paragraphs. 1C. To understand and implement basic HTML tags such as image, hyperlink, text formatting, and inline styling to design a simple webpage.	6-15	
2.	Week 2 2AvTo design a simple web page using HTML that demonstrates the use of background colour, background image, heading styling, and preformatted text.	16-19	
3.	Week-3 3A. TIMETABLE USING TABLE TAGS	20-30	
4.	Week-4 4a. a collage wants to collect student details through an online admission form. Design a registration form using only html tags (no css) with papers labels, table alignments, background colour and border. 4b. a company wants to design a webpage that displays two different forms side by side using I frames. One I frame should display a login page form and other should display a registration form.	31-36	

5.	Week-5 5a. To create a responsive card layout using HTML and CSS with three cards, each containing an image, title, and button. 5b. To design a mini website using iframe with navigation links (Home, About, Contact), where clicking on each link displays the respective content within the same page using internal or external CSS. 5c. To design an Amrita login page using HTML and CSS that displays different campuses, where clicking on each campus shows its details on the same page.	37-49	
6.	Week-6 <u>6.</u> Design a responsive e-commerce application using the grid layout. This application should contain different categories of products such as electronics, clothing, home-appliances, books. Based on the category we select corresponding products should be displayed on the sidebar	50-53	
7.	<u>WEEK-7</u> <u>7.</u> Create a Git hub account and create a repository named UIDLAB upload every week experiments, lab reports in that repository and make the repository	54-57	
8.	(8)A Aim: to create a webpage that displays a square that continuously rotates 360 degrees using CSS Animation.	58-64	

	(8)B Aim: to create a webpage that displays a ball that moves up and down continuously using CSS Animation. 8)C Aim: to create a webpage that displays a circle that moves horizontally and rotates at the same time.		
9.	(9)A Aim: to create a webpage that displays whether a number is even/odd or prime/composite or factorial of the number and is Armstrong number or not with HTML, CSS, JS.	65-72	
10.	(10)A Aim: write a Java script conditional statement to find the sign of product of three numbers using function. Display an alert box with specific sign (10)B Aim: To Sort 3 number using functions. Display on alert box to show the result. (10)C Aim: To find the biggest of 5 numbers. (10)D Aim: To iterate the form 0 to 15, for each iteration to check if the current number is odd or even, and display the message to the screen. (10)E Aim: To write a JS program to read 6 subject marks of a	73-85	

	student and then compute the average marks of the student. Then the average is used to determine the corresponding grade.		
11.	a) A website requires users to enter a username. The system should display a message while typing if the username is less than 5 characters. b) A registration form should validate email format before submission. If invalid, the form should not be submitted. c) A system should check password strength while typing and display whether it is weak or strong. d) A form should validate a mobile number when the user leaves the input field. The number must be exactly 10 digits. e) Design an HTML page, such that the registration form must validate: Name (not empty) Email (valid format) Password (minimum 6 characters)	86-95	
12	Installation of Electron framework application.	_96-101	

Week 1:

1A.To understand the installation process of a VS code and get familiar with its environment for web development.

AIM: Downloading and installation of vs code:

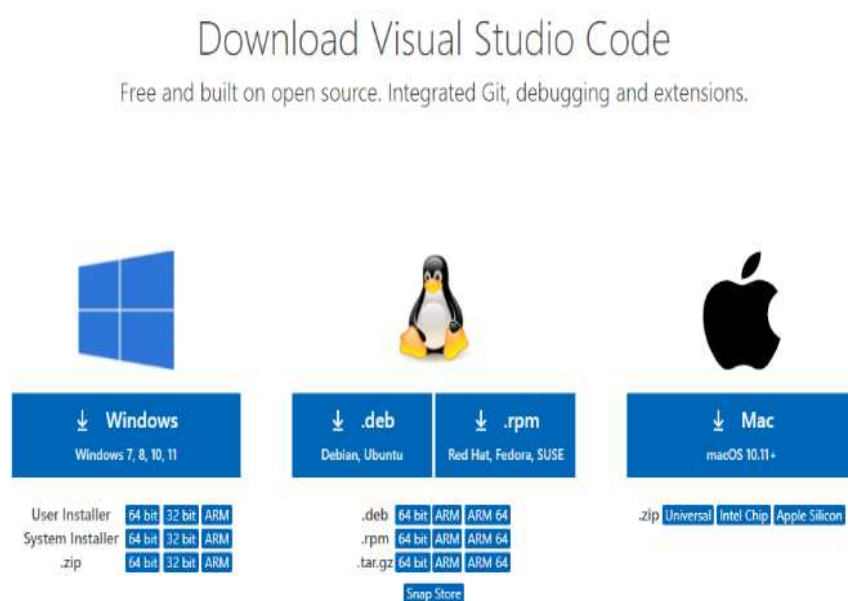
INSTALLATION:

STEP-1:

- Turn on your computer.
- Open any web browser such as Google Chrome or Firefox.
-

STEP-2:

-
- Visit the Official Website <https://code.visualstudio.com>
- On the homepage, click the **Download for Windows** button.
- The website automatically detects your operating system.
- A setup file with .exe extension will start downloading.'



STEP-3:

- Open **File Explorer**.
- Go to the **Downloads** folder.
- Find the file named similar to VSCodeUserSetup-x64.exe.

-

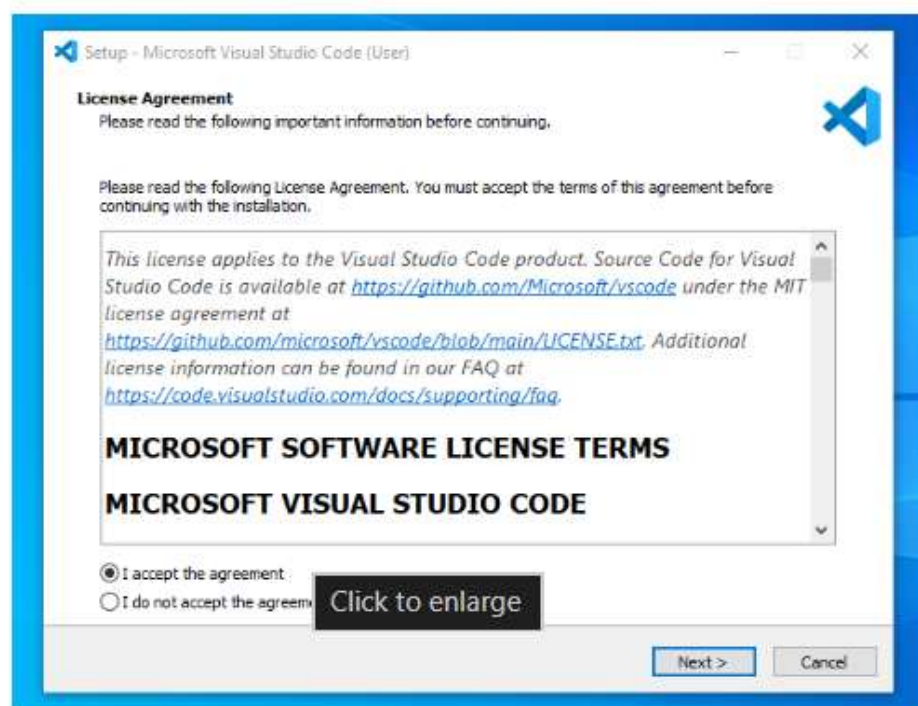
STEP-4:

- Double-click on the downloaded setup file.
- If a security prompt appears, click **Yes** to allow installation.
- The VS Code setup wizard will open.

-

STEP-5:

- Read the license terms shown on the screen.
- Select **I accept the agreement**.
- Click **Next** to continue.



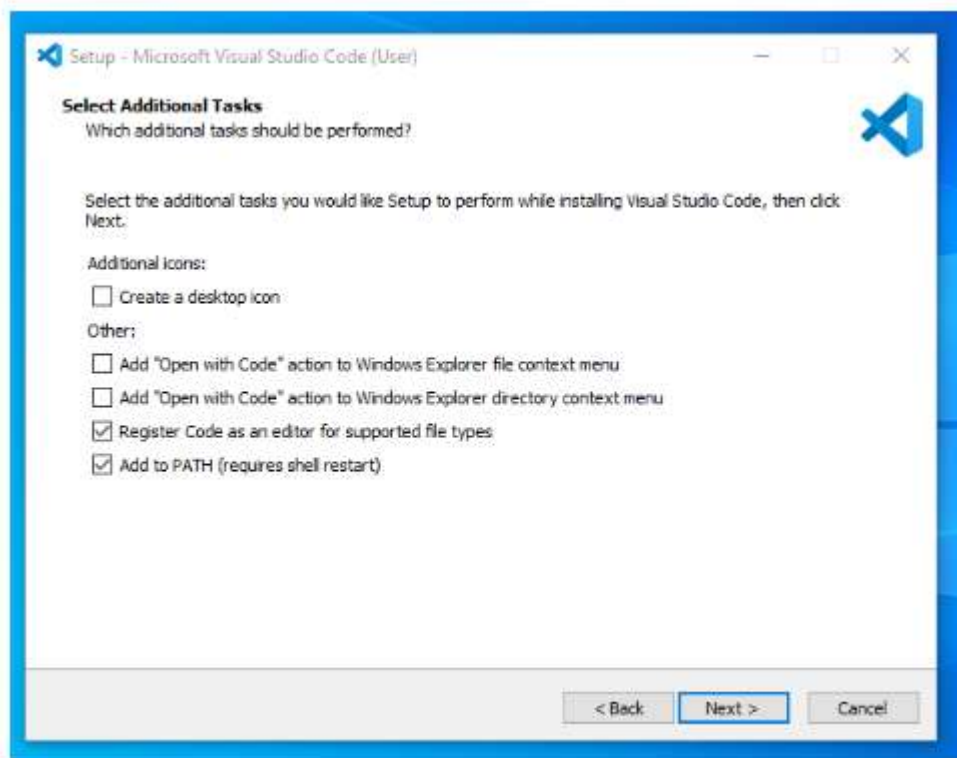
-

STEP-6:

- Choose the default installation folder.
- Click NEXT without changing the path.

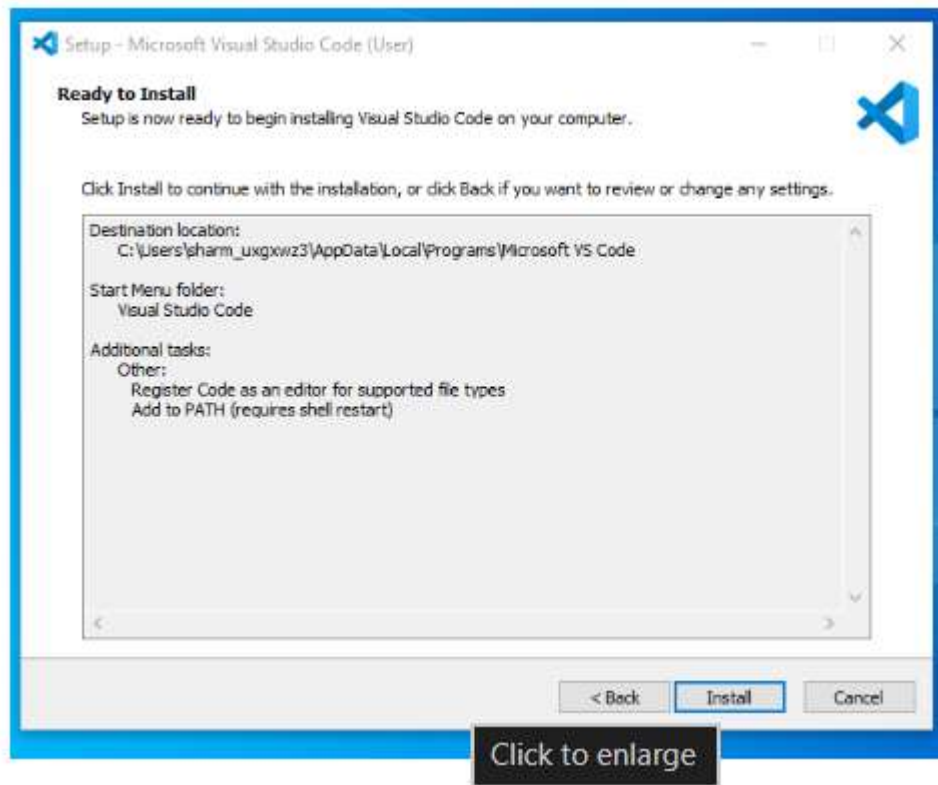
STEP-7:

- Tick the following options for better usage:
 - Add to PATH
 - Create desktop icon
 - Register Code as an editor for supported file types
- Click **Next**.



STEP-8:

- Click the **Install** button.
- Wait for the installation process to complete.
- This may take a few seconds.



-

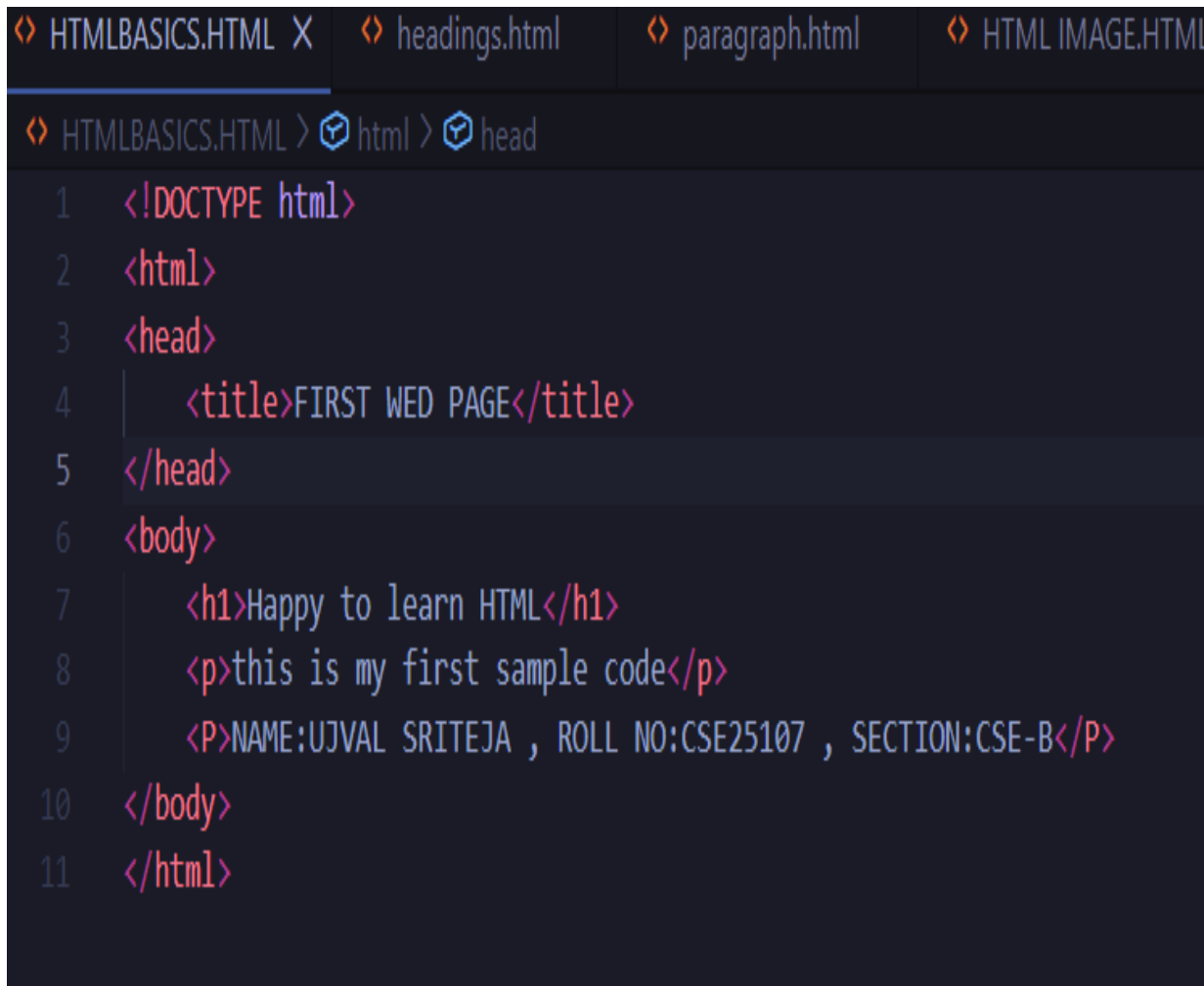
STEP-9:

- After installation, tick **Launch Visual Studio Code**.
- Click **Finish**.
- VS Code will open successfully.

1b. To learn the basic structure of an HTML document and create a simple webpage using headings and paragraphs.

AIM: STRUCTURE OF HTML

CODE:

A screenshot of a code editor with a dark theme. At the top, there are four tabs: 'HTMLBASICS.HTML X', 'headings.html', 'paragraph.html', and 'HTML IMAGE.HTML'. The 'HTMLBASICS.HTML' tab is active. Below the tabs, the breadcrumb path shows 'HTMLBASICS.HTML > html > head'. The code is as follows:

```
1 <!DOCTYPE html>
2 <html>
3 <head>
4   <title>FIRST WED PAGE</title>
5 </head>
6 <body>
7   <h1>Happy to learn HTML</h1>
8   <p>this is my first sample code</p>
9   <p>NAME:UJVAL SRITEJA , ROLL NO:CSE25107 , SECTION:CSE-B</p>
10 </body>
11 </html>
```

Above picture refers the basic structure of html

EXPLANATION:

1. DOCTYPE HTML:

- DOCTYPE HTML in above code tells that document is written in HTML.
- This line is always written at the top.

2.<HTML>:

- This is the root element of the HTML page.
- The code for preparing a webpage is written in between the opening and closing tags of HTML.

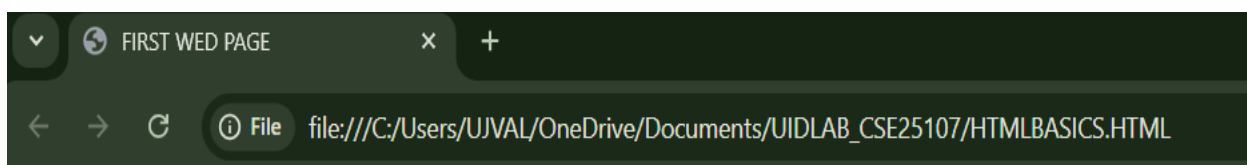
3.<HEAD>:

- IT is used to title the webpage which appears at the place of new tab instead of new tab.

4.<BODY>:

- ALL code of the main body of WEBPAGE is written in between body tags.
- <h1> and <p> tags are used for HEADING AND PARAGRAPH and many other tags are used in between body tags

OUTPUT:



Happy to learn HTML

this is my first sample code

NAME:UJVAL SRITEJA , ROLL NO:CSE25107 , SECTION:CSE-B

1c. BASIC TAGS OF HTML:

AIM: USING THE BASIC HTML TAGS

CODE:

```

1  <!DOCTYPE html>
2  <html>
3  <head>
4    <title>UNIVERSITY INFORMATION</title>
5  </head>
6  <body>
7    <h1>ABOUT UNIVERSITY</h1>
8    <h1 style="color: teal;background: -moz-repeating-radial-gradient();box-sizing: content-box;">AMRITA UNIVERSITY<br>am
9    <p title="about my self" style="background-color: darkgrey;color: darkgreen;text-box-edge: inherit;">NAME : UJVAL S
10   
11   <!--PRE=PREFORMATED TEST-->
12   <pre style="width: max-content;">
13     Amrita Vishwa Vidyapeetham (Amrita University) is a private, deemed university in India founded by Mata Amritananda
14     It has multiple campuses across India and is accredited with a high grade by the National Assessment and Accredita
15   </pre>
16   <a href="https://www.amrita.edu/campus/amaravati/">AMRITA UNIVERSITY</a>
17 </body>
18 </html>

```

EXPLANATION:

THE TAGS USED ABOVE ARE:

- <title> </title>
- <h1></h1>
- <p></p>
-
- <a>
- <pre></pre>
-

<title>:

- It is a tag used in between the tags of head.
- It is used to customize the name of the web page in place of new tab.
- It has closing tag </title>

<h1>:

- H in the tag refers to heading.
- There are six levels of headings which we can use in creating the web pages.
- They are <h1> , <h2> , <h3> , <h4> , <h5> , <h6>
- They used to change the level of heading in web pages]
- It has closing tag </h1>.

<p>:

- p- in the tag refers to the paragraph.
- <p> tag is used to create a paragraph in website.
- It has a closing tag </p>.
- It doesn't care about the gaps and give the direct paragraphs without any break.

<image>:

- Image tag is used to add a image to our creating website.
- In this tag we have **alt** and it is used for alternate text if any problem in loading the image.
- This tag doesn't have closing tag.

<a>:

- a tag is used to add a links in this our website.
- It contains href which is a part to add link and we can give a name that refers link as shown in above code.
- It has closing tag.

<pre>:

- Pre tag is used to add the lines of paragraph separately as shown in above code.
- Pre refers to PREFORMATED TEXT.
- It has closing tag </pre>.

**
:**

- Br tag is used to break the continuous flow of sentence while using <p> tag.
- It will break the paragraph.
- It doesn't have closing tag.

OUTPUT:



ABOUT UNIVERSITY

AMRITA UNIVERSITY amaravati

NAME : UJVAL SRITEJA , ROLL NUMBER : CSE25107 , SECTION : CSE-B



Amrita Vishwa Vidyapeetham (Amrita University) is a private, deemed university in India founded by Mata Amritanandamayi Devi ("Amma"). It has multiple campuses across India and is accredited with a high grade by the National Assessment and Accreditation Council (NAAC).

[AMRITA UNIVERSITY](#)

WEEK 2:

2A. To design a simple web page using HTML that demonstrates the use of background colour background image, heading styling, and preformatted text.

Aim: To design a simple web page using HTML that demonstrates the use of background colour background image, heading styling, and preformatted text.

PROGRAM:

```

<!DOCTYPE HTML>
<html>
  <head>
    <title></title>
  </head>
  <body style="background-color: #f0f0f0; background-image: url('https://images.shutterstock.com/image-data/images/175997782/894655.jpg'); background-repeat: no-repeat; background-size: 100% 100%; border: 1px solid black;">
    <pre style="text-align: center;">
      <h1 style="color: #000080; font-style: italic; font-size: x-large; font-style: oblique; font-size: xx-large;"><h1>AMRITA</h1>
    </pre>
    <p style="font-size: larger; text-align: center;">Amrita Vishwa Vidyapeetham is a leading multi-disciplinary and research-intensive university, established amidst the scenic beauty of the Krishna River and surrounded by natural wonders and historical sites, Amravati city embodies the principles of sustainability and livability, we are building one of the most buoyant centre of the research, education and technological advancement, in the all new greenfield capital city, Uli>
    <table border="1" cellpadding="5" cellspacing="5" width="50%" align="center" style="background-color: #f0f0f0;">
      <caption style="text-align: center; font-size: x-large; font-weight: bold; color: #d3d3d3;">AMRITA STUDENTS DETAILS</caption>
      <tr>
        <th>
          STUDENT NAME
        </th>
        <th>
          COURSE
        </th>
        <th>
          SECTION
        </th>
        <th>
          CONTACT
        </th>
      </tr>
      <tr>
        <td>E.0796</td>
        <td rowspan="4">CSE</td>
        <td>CSE-8</td>
        <td>9400790790</td>
      </tr>
    </table>
  </body>
</html>

```



```

<table border="1" cellpadding="5" cellspacing="5" width="50" align="center" style="background-color: #f0f0f0; border-collapse: collapse; margin: 20px auto;">
<tr>
<td colspan="2" style="background-color: #f0f0f0; text-align: center; padding: 5px;">
CONTACT
</td>
</tr>
<tr>
<td style="background-color: #f0f0f0; padding: 5px;">
NAME:
</td>
<td style="background-color: #f0f0f0; padding: 5px;">
<input type="text" value="<!-->"/>
</td>
</tr>
<tr>
<td style="background-color: #f0f0f0; padding: 5px;">
ADDRESS:
</td>
<td style="background-color: #f0f0f0; padding: 5px;">
<input type="text" value="<!-->"/>
</td>
</tr>
<tr>
<td style="background-color: #f0f0f0; padding: 5px;">
PHONE:
</td>
<td style="background-color: #f0f0f0; padding: 5px;">
<input type="text" value="<!-->"/>
</td>
</tr>
<tr>
<td style="background-color: #f0f0f0; padding: 5px;">
E-MAIL:
</td>
<td style="background-color: #f0f0f0; padding: 5px;">
<input type="text" value="<!-->"/>
</td>
</tr>
<tr>
<td colspan="2" style="background-color: #f0f0f0; padding: 5px; text-align: center;">
<input type="button" value="Submit" />
</td>
</tr>
</table>

```

- style attribute – Used to apply CSS styles directly to an element to control colour, alignment, background, font size, spacing, borders, and layout.
- Style tag is to design the created web page
- There are many attributes in style tag which is used to design the web page

OUTPUT:

AMRITA VISHWA VIDHYAPEETHAM

ABOUT AMRITA

Amrita Vishwa Vidyapeetham is a leading multi-disciplinary and research-intensive university, establishing its campus in Amarakali. Nestled amidst the scenic beauty of the Krishna River and surrounded by natural wonders and historical sites, Amarakali city embodies the principles of sustainability and livability. We are building one of the most buoyant centre of the research, education and technological advancement, in the all new greenfield capital city.

AMRITA STUDENTS DETAILS

STUDENT NAME	COURSE	SECTION	CONTACT
E.UJVAL	CSE	CSE B	9493390790
M.GOKUL		CLASS TOPPER	
P.VARUN		CSE B	9960863853
P.BHASKAR		CSE B	8123466611

TAGS USED:

- `<u>` – Underlines the enclosed text to give it visual emphasis.
- `<b>` – **Displays text in bold without indicating extra importance.**
- `table` – Creates a table to arrange data in rows and columns.
- `tr` – Defines a table row inside the table.
- `td` – Defines a normal table data cell used to store text, images, or other content.
- `Th` – Defines a table header cell. It 3
- is bold and centred by default and is used for headings.
- `style` attribute – Used to apply CSS styles directly to an element to control colour, alignment, background, font size, spacing, borders, and layout.
- `Colour`: Colour attribute in style used to decide the colour of text.
- `Colspan`: `colspan` is an attribute used to combine the columns.
- `Rowspan`: `rowspan` is an attribute used to combine the rows.
- `Background colour`: `background colour` is an attribute is used to set the background colour of the webpage.
- `Background image`: `background image` is an attribute is used to set the background image of the webpage.
- `Font style`: `font style` is an attribute used to change the font of the text in webpage.
- `Text align`: `text align` is an attribute used to set the text at centre of the webpage.
- `Cellpadding` and `cell spacing` is used to set the borders to the table.
- `Font size`: `font size` is an attribute used to change the size of the text in webpage.

WEEK-3

3A. To design and develop an academic timetable using HTML tables
Applying row span, column span, and styling for proper structure and presentation.

program:

```

1 <!DOCTYPE html>
2 <html>
3 <head>
4 <title>TIME TABLE</title>
5 </head>
6 <body>
7 <div>NAME: E.UOVAL SRITEJA , CLASS-CSE-B , ROLL NUMBER :
8 </div>
9 <table border="1" cellpadding="3" cellspacing="1" width="85%" align="center">
10 <tr>
11 <tr>
12 <td colspan="12" style="text-align: center;"><b>MASTER TIME TABLE CSE II SEMESTER B SECTION</b></td>
13 </tr>
14 <tr>
15 <td colspan="4"><b>Dept.cse</b></td>
16 <td colspan="8" rowspan="3"><b>CLASSROOM VENUE : 511</b></td>
17 </tr>
18 <tr>
19 <td colspan="4"><b>Class : B.tech. CSE-B</b></td>
20 </tr>
21 <tr>
22 <td colspan="4"><b>Semester - II</b></td>
23 </tr>
24 <tr>
25 <td colspan="4"><b>Class Advisors:</b>
26 <div>1. Name: Dr. Karmala Gouri Raghavendra Narayan
27 Email : kgr_narayan@v.sarita.edu
28 Mobile Number : +91-9785489393</div>
29 </td>
30 <td colspan="8"><b>2. Name: Dr. Ankit Sachan</b>
31 <div>Email : s_ankit@v.sarita.edu
32 Mobile Number : +91-8381880129</div>
33 </td>
34 </tr>
35 <tr>
36 <th>TIME/DAY</th>
37 <th>
38 <th>SLOT-1</th>
39 <th>
40 <th>SLOT-2</th>
41 <th>

```

```

1 <html>
2 <body>
3   <table border="1" cellpadding="5" cellspacing="1" width="85%" align="center" >
4     <tr>
5       </td>
6       <td>
7         SLOT-1
8       </td>
9       <td>
10        SLOT-2
11      </td>
12      <td>
13        10:25-10:45
14      </td>
15      <td>
16        SLOT-3
17      </td>
18      <td>
19        SLOT-4
20      </td>
21      <td>
22        SLOT-5
23      </td>
24      <td>
25        1:15-2:10
26      </td>
27      <td>
28        SLOT-6
29      </td>
30      <td>
31        3:00 - 3:10
32      </td>
33      <td>
34        SLOT-7
35      </td>
36      <td>
37        SLOT-8
38      </td>
39    </tr>
40    <tr>
41      <td>
42        MONDAY
43      </td>
44      <td colspan="2" > LINEAR ALGEBRA </td>

```

```

1 <html>
2 <body>
3   <table border="1" cellpadding="5" cellspacing="1" width="85%" align="center" >
4     <tr>
5       <td>
6         MONDAY
7       </td>
8       <td colspan="2" > LINEAR ALGEBRA </td>
9       <td rowspan="5" style="background-color: yellow; writing-mode: vertical-lr; text-align: center;">SHORT BREAK</td>
10      <td>GLIMPSE OF GLORIOUS</td>
11      <td>MODERN PHYSICS</td>
12      <td>LIBRARY</td>
13      <td rowspan="5" style="background-color: yellow; writing-mode: vertical-lr; text-align: center;"> LUNCH BREAK</td>
14      <td colspan="2" >MANUFACTURING PRACTICE LAB</td>
15      <td rowspan="5" style="background-color: yellow; writing-mode: vertical-lr; text-align: center;"> SHORT BREAK</td>
16      <td colspan="2" >MANUFACTURING PRACTICE LAB</td>
17    </tr>
18    <tr>
19      <td>TUESDAY</td>
20      <td>DISCRETE MATHEMATICS</td>
21      <td>LINEAR ALGEBRA</td>
22      <td>MODERN PHYSICS</td>
23      <td>OBJECT ORIENTED PROGRAMMING</td>
24      <td>USER INTERFERENCE DESIGN</td>
25      <td>GLIMPSE OF GLORIOUS</td>
26      <td>SPORTS/LIBRARY</td>
27    </tr>
28    <tr>
29      <td>WEDNESDAY</td>
30      <td colspan="2" >OBJECT ORIENTED PROGRAMMING LAB</td>
31      <td>DISCRETE MATHEMATICS</td>
32      <td>USER INTERFERENCE DESIGN</td>
33      <td>GLIMPSE OF GLORIOUS</td>
34      <td>LIBRARY/SPORTS</td>
35      <td>LIBRARY/SPORTS</td>
36    </tr>
37    <tr>
38      <td>THURSDAY</td>
39      <td>LINEAR ALGEBRA</td>
40      <td>OBJECT ORIENTED PROGRAMMING</td>
41      <td>DISCRETE MATHEMATICS</td>
42      <td style="color: red;">COUNSELLING</td>
43      <td>LIBRARY</td>
44      <td>MODERN PHYSICS</td>
45      <td>LIBRARY/SPORTS</td>

```

```

<html>
<body>
<table border="1" cellpadding="5" cellspacing="1" width="85%" align="center">
<tr>
<td>LIBRARY</td>
<td colspan="2">MODERN PHYSICS</td>
<td colspan="2">LIBRARY/SPORTS</td>
</tr>
<tr>
<td colspan="4">FRIDAY</td>
<td colspan="2">USER INTERFACE DESIGN LAB</td>
<td colspan="2">LINEAR ALGEBRA</td>
<td colspan="2">OBJECT ORIENTED PROGRAMMING</td>
<td colspan="2">LIBRARY/SPORTS</td>
<td style="color: red;" colspan="2">COURSE LING</td>
<td colspan="2">DISCRETE MATHEMATICS LAB</td>
</tr>
<tr>
<td colspan="4">Course Code</td>
<td colspan="2"><b></td>
<td colspan="2">Course Name</td>
<td colspan="2"></td>
<td colspan="2"></td>
<td colspan="2">L-T-P</td>
<td colspan="2">Credits</td>
<td colspan="2">Faculty Name</td>
<td colspan="2"></td>
<td colspan="2">Assisting Facult</td>
</tr>
<tr>
<td colspan="4">22MAT117</td>
<td colspan="2">Discrete Mathematics</td>
<td colspan="2"></td>
<td colspan="2">3-0-2</td>
<td colspan="2">4</td>
<td colspan="2">Dr. Srivivasa Rao Thota</td>
<td colspan="2"></td>
<td colspan="2"><b></td>
</tr>
<tr>
<td colspan="4">22MAT117</td>
<td colspan="2">Linear Algebra</td>

```

```

<html>
<body>
<table border="1" cellpadding="5" cellspacing="1" width="85%" align="center">
<tr>
<td colspan="2"><b></td>
</tr>
<tr>
<td colspan="2">22CSE111</td>
<td colspan="2">Object Oriented Programming</td>
<td colspan="2"></td>
<td colspan="2">3-0-2</td>
<td colspan="2">4</td>
<td colspan="2">Dr. Balaji KK</td>
<td colspan="2"></td>
<td colspan="2"><b></td>
</tr>
<tr>
<td colspan="2">22CSE111</td>
<td colspan="2">Object Oriented Programming</td>
<td colspan="2"></td>
<td colspan="2">3-0-2</td>
<td colspan="2">4</td>
<td colspan="2">Dr. Balaji KK</td>
<td colspan="2"></td>
<td colspan="2"><b></td>
</tr>
<tr>
<td colspan="2">22PHY115</td>
<td colspan="2">Modern Physics</td>
<td colspan="2"></td>
<td colspan="2">2-1-0</td>
<td colspan="2">3</td>
<td colspan="2">Dr. C H Kesavulu</td>
<td colspan="2"></td>
<td colspan="2"><b></td>
</tr>
<tr>
<td colspan="2">22CSE111</td>
<td colspan="2">User Interface Design</td>
<td colspan="2"></td>
<td colspan="2">2-0-2</td>
<td colspan="2">3</td>
<td colspan="2">Dr. Rajkumar Batches</td>
<td colspan="2"></td>
<td colspan="2"><b></td>

```

```

1  <html>
2  <body>
3      <table border="1" cellpadding="5" cellspacing="1" width="85%" align="center" >
4          <tr>
5              <td><b> 2-0-2</b></td>
6              <td><b>3</b></td>
7              <td colspan="3"><b> Dr.Rajkumar Satchu</b></td>
8              <td></td>
9              <td colspan="2"><b>Mr. Barge Bharadwaj</b></td>
10         </tr>
11         <tr>
12             <td><b>23ME115</b></td>
13             <td colspan="2"><b>Manufacturing Practice</b></td>
14             <td></td>
15             <td><b> 8-0-3</b></td>
16             <td><b>1</b></td>
17             <td colspan="3"><b> Dr. Sathies S.</b></td>
18             <td></td>
19             <td colspan="2"><b></b></td>
20         </tr>
21         <tr>
22             <td><b>22A0111</b></td>
23             <td colspan="2"><b>Glimpses of Glorious India</b></td>
24             <td></td>
25             <td><b> 2-0-1</b></td>
26             <td><b>2</b></td>
27             <td colspan="3"><b> Dr. Siva Panuganti</b></td>
28             <td></td>
29             <td colspan="2"><b></b></td>
30         </tr>
31         <tr>
32             <td></td>
33             <td colspan="2"><b>Total (15L + 1Tut +12P =28 hrs)</b></td>
34             <td></td>
35             <td><b>15-1-12</b></td>
36             <td><b>21</b></td>
37             <td colspan="3"></td>
38             <td></td>
39             <td colspan="2"></td>
40         </tr>
41     </table>
42 </body>
43 </html>

```


OUTPUT:NAME: UVAAS SRITEJA, CLASS: CSE-B, ROLL NUMBER:

MASTER TIME TABLE CSE II SEMESTER B SECTION											
Dept.cse			CLASSROOM VENUE : 511								
Class :B.tech CSE-B											
Semester - II											
Class Advisors: 1. Name: Dr. Karmala Gowri Engalarendra Narayan Email : kgp_narasimhas@amrita.edu Mobile Number : +91-9706408383			2. Name: Dr. Ankit Sachan Email : s_ankit@amrita.edu Mobile Number : +91-981601129								
TIME/DAY	SLOT-1	SLOT-2	1045-1045	SLOT-3	SLOT-4	SLOT-5	115-230	SLOT-6	240-240	SLOT-7	SLOT-8
MONDAY	LINEAR ALGEBRA		MIDCARE LIBRARY	GLIMPSE OF GLORIOUS	MODERN PHYSICS	LIBRARY	LUNCH 1 HOUR 45	MANUFACTURING PRACTICE LAB	MIDCARE LIBRARY	MANUFACTURING PRACTICE LAB	
TUESDAY	DISCRETE MATHEMATICS	LINEAR ALGEBRA		MODERN PHYSICS	OBJECT ORIENTED PROGRAMMING	USER INTERFERENCE DESIGN		GLIMPSE OF GLORIOUS		SPORTS LIBRARY	
WEDNESDAY	OBJECT ORIENTED PROGRAMMING LAB			DISCRETE MATHEMATICS	USER INTERFERENCE DESIGN	GLIMPSE OF GLORIOUS		LIBRARY SPORTS		LIBRARY SPORTS	
THURSDAY	LINEAR ALGEBRA	OBJECT ORIENTED PROGRAMMING		DISCRETE MATHEMATICS	COUNSELLING	LIBRARY		MODERN PHYSICS		LIBRARY SPORTS	
FRIDAY	USER INTERFERENCE DESIGN LAB			LINEAR ALGEBRA	OBJECT ORIENTED PROGRAMMING	LIBRARY SPORTS		COUNSELLING		DISCRETE MATHEMATICS LAB	
Course Code	Course Name			L-T-P	Credits	Faculty Name		Assisting Faculty			
23MAT117	Discrete Mathematics			3-0-2	4	Dr. Srivirasa Rao Thota					
23MAT117	Linear Algebra			3-0-2	4	Mr.Sagarot Rinzapathi					
23CSE111	Object Oriented Programming			3-0-2	4	Dr. Balaji TK					
23CSE111	Object Oriented Programming			3-0-2	4	Dr. Balaji TK					
23PHY115	Modern Physics			2-1-0	3	Dr. C R Kesavulu					
23CSE113	User Interface Design			2-0-2	3	Dr.Rajkumar Bateha		Mr. Ranga Bharadwaj			
23MEP115	Manufacturing Practice			0-0-3	1	Dr. Saritha S.					
23ADM111	Glimpses of Glorious India			2-0-1	2	Dr. Siva Panaganti					
	Total (SGL + TUT + SLP = 26 hrs)			15-1-12	21						

TAGS USED:

- `<u>` – Underlines the enclosed text to give it visual emphasis.
- `<b>` – **Displays text in bold without indicating extra importance.**
- `table` – Creates a table to arrange data in rows and columns.
- `tr` – Defines a table row inside the table.
- `td` – Defines a normal table data cell used to store text, images, or other content.
- `Th` – Defines a table header cell. It is bold and centred by default and is used for headings.
- `style` attribute – Used to apply CSS styles directly to an element to control colour, alignment, background, font size, spacing, borders, and layout.
- `Colour`: `Colour` attribute in `style` used to decide the colour of text.
- `Colspan`: `colspan` is an attribute used to combine the columns.
- `Rowspan`: `rowspan` is an attribute used to combine the rows.
- `Font style`: `font style` is an attribute used to change the font of the text in webpage.
- `Text align`: `text align` is an attribute used to set the text at centre of the webpage.
- `Cellpadding` and `cell spacing` is used to set the borders to the table.
- `Font size`: `font size` is an attribute used to change the size of the text in webpage.

3b. To create a comprehensive webpage that includes an academic timetable along with syllabus representation using images and structured HTML elements.

Aim: To create a comprehensive webpage that includes an academic timetable along with syllabus representation using images and structured HTML elements.

Program

```

1 <!DOCTYPE html>
2 <html lang="en">
3 <head>
4   <title>COURSE DETAILS</title>
5 </head>
6 <body>
7   <h1 style="text-align: center; color: #deeppink;"><u>AMRITA VISHVA VIDHYAPEETAM</u></h1>
8   <h2 style="color: #deeppink; border: 3px solid black; text-align: center;">DETAILS ABOUT COURSES</h2>
9
10  <h3 style="color: #deeppink;"><u>ABOUT COURSES</u></h3>
11
12  <div><h2 style="color: #deeppink;"><u>CS23CSE113 User Interface Design 2-0-2 : </u></h2>
13    <pre style="font-size: x-large;"><b>
14      <h4><u style="color: #slategray;">Course Objectives : </u></h4>
15
16      The course focuses on the basic understanding of user interface design by applying HTML,
17      CSS and Java Script. The course introduces the necessary skills to develop web applications
18      with simple animation and transitions. An overview of UI frameworks is given. Working of
19      Internet is discussed. The protocols required are introduced.
20
21      <h4><u style="color: #slategray;">Course Outcomes : </u></h4>
22
23      CO1: To understand the basics and working of World Wide Web
24
25      CO2: To understand the fundamentals of HTML5
26
27      CO3: To understand the fundamentals of CSS and Java Script
28
29      CO4: To design and deploy a single web application</b>
30    </pre></div>
31    <br>
32    <br>
33
34  <div><h2 style="color: #deeppink;"><u>CS23MAT116 Discrete Mathematics 3-0-2 </u></h2>
35    <pre style="font-size: x-large;">
36      <b>

```

```
h4<u style="color: #slategrey;">Course Objectives</u></h4>
```

```
understand the logic and various functions.  
understand the basic concept of combinatorics.  
understand the concepts of recurrence relations and their applications.  
understand the concepts of equivalence and partial order relations.  
understand various definitions and theorems on graph theory.
```

```
h4<u style="color: #slategrey;">Course Outcomes</u></h4>
```

```
O1: To understand the basic concepts of Mathematical reasoning and basic counting techniques.  
O2: To understand the recursive functions and apply the concepts of generating functions to solve the recurrence relations.  
O3: Apply the concepts of divide and conquer method and principle of inclusion and exclusion to solve some simple algorithms in discrete mathematics.  
O4: To understand the concepts of various types of relations, partial ordering and equivalence relations.  
O5: To understand the basic concepts of graph theory and apply to shortest path problems.
```

```
</b>
```

```
</pre>
```

```
</div>
```

```
<br>
```

```
<br>
```

```
<div>h2 style="color: #deeppink;"><u>23MAT117 Linear Algebra 1-0-2</u></h2>
```

```
pre style="font-size: x-large;">h3
```

```
h4<u style="color: #slategrey;">Course Objectives</u></h4>
```

```
understand the basic concepts of vector space, subspace, basis and dimension. Also to understand the orthogonality concepts and apply to various problems.
```

```
h4<u style="color: #slategrey;">Course Outcomes</u></h4>
```

```
O1: To understand the basic concepts of vector space, subspace, basis and dimension.
```

```
O2: To understand the basic concepts of inner product space, norm, angle, Orthogonality and projection and implementing the Gram-Schmidt process, etc.
```

```

pre style="font-size: x-large;"><h3>
O-PD Mapping:</pre></div>

<br>
<br>

<div><h2 style="color: #deeppink;"><u>23CSE111 Object Oriented Programming 3-0-2 </u></h2>
pre style="font-size: x-large;">
<u><h4 style="color: #slategrey;">Course Objectives</h4></u>

he course aims at teaching students to develop Object-Oriented software using the Unified Modelling Language and the Java Programming Language to f
his course motivates the students to think of problem solving in an Object-Oriented way using the methods and tools that support the paradigms.

<h4 style="color: #slategrey;"><u>Course Outcomes</u></h4>

O1: Understand Object Oriented paradigms and represent the problem using objects and classes.

O2: Apply the Object Oriented concepts to design and develop effective models using UML.

O3: Develop program logic in Java from design models in UML.

O4: Design applications with procedural and data abstraction using Java libraries</pre>
</div>

<br>
<br>

<div><h4><u style="color: #slategrey;">Course Outcomes</u></h4>
pre style="font-size: x-large;">
O01: Practice safety procedures in a shop floor environment.

O2: Select appropriate tools and methods for basic manufacturing processes.

O3: Build simple geometries using additive manufacturing process.

```

```

<div style="color: #deeppink; border: 1px solid black; text-align: center; padding: 5px;">
</div>

<br>
<br>

<div><h2 style="color: #deeppink;"><u>22A0PM11: Glimpses of Glorious India 2-8-1</u></h2>
<h3 style="color: #slategrey;">Course Objectives</h3>
<pre style="font-size: x-large;">The course aims at introducing Bhārath inshell to the student, which includes the sources of Indian thoughts,
    who shaped various disciplines, India's nutfignificant contribution to the man kind, the current stature of Indian in the geopolitics and
    Indian approach to science and ecology.</pre>

<h4><u style="color: #slategrey;">Course Outcomes</u></h4>
<pre style="font-size: x-large;">
    CO1: Will be able to recognise the call of Upanishads and outstanding personalities for confronting the wicked in the real world while admiring
        the valour, pursuit and divinity in both classical and historical female characters of India.

    CO2: Will get introduced to Acharya Chanakya, his works, and his views on polity and nation to find synchrony between public and personal life,
        alongside understanding India's cultural nuances and uniqueness concerning the comprehension of God across major global communities.

    CO3: Will be able to appreciate Bhagavad Gita as the source of the Indian worldview through the various Yogic lessons enshrined in it, making it
        one of India's numerous soft powers, and also understand the faith-oriented mechanism of preserving nature.

    CO4: Will be informed about the enormous contribution of Indian civilisation over two and a half millennia to humanity and develop awareness about
        India's approach toward science, devoid of dogmas and rooted in humanism.
</pre>
</div>

<div>
    <p><a href="file:///C:/Users/UJVAL/OneDrive/Documents/UIDLAB_CSE25107/timetable.html"> click here for timetable</a></p>
</div>
</div>

```

Output

The screenshot displays a web browser window with the URL `file:///C:/Users/Vijay/Desktop/AMBITA_VISHWA_VIDHYAPEETAM_CSE25107/webpage%20of%20courses.html`. The page title is "AMBITA VISHWA VIDHYAPEETAM" and the main heading is "DETAILS ABOUT COURSES".

ABOUT COURSES

ECECE113 User Interface Design 2-0-2

Course Objectives :

The course focuses on the basic understanding of user interface design by applying HTML, CSS and Java Script. The course introduces the necessary skills to develop web applications with simple animation and transitions. An overview of UI Frameworks is given. Working of Internet is discussed. The pre-requisite required are listed below.

Course Outcomes :

CO0: To understand the basic and working of World Wide Web
 CO0: To understand the fundamentals of HTML5
 CO0: To understand the fundamentals of CSS and Java Script
 CO0: To design and deploy a simple web application

EEMATE102 Discrete Mathematics 3-0-2

Course Objectives :

Understand the logic and various functions.
 Understand the basic concept of combinatorics.
 Understand the concepts of recurrence relations and their applications.
 Understand the concepts of optimization and partial order relations.
 Understand various definitions and theorems on graph theory.

Course Outcomes :

The course aims at teaching students to develop Object Oriented software using the Unified Modeling Language and the Java Programming Language to first year students. This course motivates the students to think of problem solving in an Object Oriented way using the methods and tools that support the paradigm.

Course Outcomes

CO1: Understand Object Oriented paradigm and represent the problem using objects and classes.
 CO2: Apply the Object Oriented concepts to design and develop effective models using UML.
 CO3: Develop program logic in Java from design models in UML.
 CO4: Design applications with procedural and data abstractions using Java Streams

Course Outcomes

CO1: Practice safety procedures in a shop floor environment.
 CO2: Select appropriate tools and methods for basic manufacturing processes.
 CO3: Build simple mechanisms using additive manufacturing process.
 CO4: Perform basic metal joining using welding and soldering.
 CO5: Design, simulate and test simple pneumatic and electric pneumatic circuit for automation applications.
 CO6: Understand the functioning of mechatronic systems and realize the importance of recent developments in mechatronic technologies.

E2ADM111 Glories of Glorious India 2-0-2

Course Objectives

The course aims at introducing Bharat's culture to the students, which includes the essence of Indian thoughts, essence presentation of the shaped various disciplines, India's indigenous contribution to the world, the current status of India in the progression and Indian approach to science and technology.

Course Outcomes

CO1: Will be able to recognize the call of Upanishads and outstanding personalities for transcending the wicked in the real world while abiding the values, pursuit and identity in both classical and historical facets character of India.
 CO2: Will get introduced to Acharya Chanakya, his works, and his views on polity and culture to find synchrony between public and personal life, alongside understanding India's cultural moorings and tenets concerning the comprehension of God across major global religions.
 CO3: Will be able to appreciate Bhagavad Gita as the source of the Indian worldview through the various Yugas lessons enshrined in it, making it one of India's numerous soft powers, and also understand the birth-oriented mechanism of preserving nation.
 CO4: Will be informed about the enormous contribution of Indian civilization over two and a half millennia to humanity and develop awareness about India's approach toward science, belief of science and need to be innovative.

[Click here for details](#)

Tags used:

- **<h1> to <h6>:** this tag is used for headings.
- **<u>:** this tag is used for underline the text in webpages.
- **:** this tag is used for bold the sentence.
- **<i>:** this tag is used for change the font into italic format.
- **<div>:** creates a section or division in webpage.
- **<pre>:** display the text in preformatted form.
- **
:** to break the sentence.
- **<p>:** for creating paragraph.
- **<a>:** create a link to access another webpage or site.
- **style:** used to decorate the webpage.

WEEK-4

4a. a collage wants to collect student details through an online admission form. Design a registration form using only html tags (no css) with papers labels, table alignments, background colour and border.

Aim: a collage wants to collect student details through an online admission form. Design a registration form using only html tags (no css) with papers labels, table alignments, background colour and border.

Program:

```

1 <!DOCTYPE html>
2 <html lang="en">
3 <head>
4 <title>COLLAGE REGISTRATION FORM</title>
5 </head>
6 <body style="background-image:url(2.jpg); background-repeat: no-repeat; background-size: cover;">
7 <form style="text-align: center;">
8 <center>
9 <table border="black" cellpadding="16px" cellspacing="1px" style="height: 300px; width: 600px; margin: 100px auto; background-color: rgba(25
10 <tr>
11 <td colspan="2" style="text-align: center; color: ■white;"> COLLAGE ADMISSION REGISTRATION</td>
12 </tr>
13 <tr>
14 <td>
15 <label for="fname" style="color: ■white;">Full Name : </label></td>
16 <td>
17 <input type="text" id="fname" placeholder="first name" style="background-color: □black; color: ■white;"><br></td>
18 </tr>
19 <tr>
20 <td><label for="email" style="color: ■white;">email :</label></td>
21 <td><input type="email" id="email" placeholder="email" style="background-color: □black; color: ■white;"></td>
22 </tr>
23 <tr>
24 <td><label for="phone number" style="color: ■white;">phone number :</label></td>
25 <td><input type="text" id="phone number" placeholder="phone number" style="background-color: □black; color: ■white;"></td>
26 </tr>
27 <tr>
28 <td>
29 <label for="gender" style="color: ■white;">Gender :</label></td>
30 <td>
31 <input type="radio" id="gender" name="gender" value="male" style="background-color: □black; color: ■white;">male
32 <input type="radio" id="gender" name="gender" value="female" style="background-color: □black; color: ■white;">female
33 <input type="radio" id="gender" name="gender" value="other" style="background-color: □black; color: ■white;">other
34 </td>
35 </tr>
36 <tr>
37 <td><label for="course" style="color: ■white;">Course To Apply :</label></td>
38 <td>
39 <select name="course" id="course" style="background-color: □black; color: ■white;">
40 <option disabled selected>select</option>
41 <option>ITECH CSE</option>
42 <option>ITECH ICE</option>
43 <option>ITECH AIDS</option>
44 <option>ITECH AIMS</option>

```


[illegible]

Output:

COLLEGE ADMISSION REGISTRATION

Full Name :

Email :

Phone Number :

Gender : ☒ Male ☐ Female ☐ Other

Course To Apply :

Hobbies : ☒ MUSIC ☒ SPORTS ☒ READING

Date Of Birth :

Others :

Address :

Notes Your Address :

Tags used:

- `<form>`: this tag is used to create a form using different attributes in this form tag.
- `<center>`: this tag is used to shift the content or form into centre of the web page.
- `<table>`: this tag is used to create a table with rows and columns in our own webpages.
- `<tr>`: for rows in table
- `<td>`: for columns in table.
- `<label>`: provides connection with label and input field and provides the label for the input field.
- `<input>`: creates a field for collect the user details in webpage.
- `<select>`: this tag is used to create a dropdown list for user preferences.
- `<option>`: this tag is used to display options to user and to collect the information from the user.
- `<text area>`: creates a multiline field for user input.
- `<button>`: creates button in webpages.
- `Type="name"`: attribute used for collect name from user.
- `Type = "email"`: attribute used for collect mail from user.
- `Type= radio`: attribute used for provide options for user.
- `Type= date`: attribute used for collect date from user.
- `Type = file`: attribute used for collect file from user.
- `Type= reset`: attribute used to create reset button in webpage.

4b. a company wants to design a webpage that displays two different forms side by side using I frames. One I frame should display a login page form and other should display a registration form.

Aim: a company wants to design a webpage that displays two different forms side by side using I frames. One I frame should display a login page form and other should display a registration form.

Program:

Login page:

```

> form 4b1.html > html > body > CENTER
1  <!DOCTYPE html>
2  <html lang="en">
3  <head>
4      <title>I FRAMES </title>
5  </head>
6  <body>
7      <CENTER>
8          <table border="black">
9              <caption style="text-align: center;">User Login</caption>
10             <tr>
11                 <th>
12                     <label for="uname"><i>UserName :</i></label>
13                 </th>
14                 <th><input type="name" id=uname</th>
15             </tr>
16             <tr>
17                 <td><label for=pass><i>PassWord :</i></label></td>
18                 <td><input type="password" id="pass"></td>
19             </tr>
20             <tr>
21                 <td colspan="2" style="text-align: center;"><button>login</button></td>
22             </tr>
23         </table>
24     </CENTER>
25
26
27
28 </body>
29 </html>

```

Registration page:

```

1 <!DOCTYPE html>
2 <html lang="en">
3 <head>
4 <title>I FRAMES</title>
5 </head>
6 <body>
7 <div>
8 <table border="black">
9 <caption style="text-align: center;">User Registration</caption>
10 <tr>
11 <th>
12 <label for="uname"><i>Full Name :</i></label>
13 </th>
14 <th><input type="text" id="uname"></th>
15 </tr>
16 <tr>
17 <td><label for="email"><i>Email :</i></label></td>
18 <td><input type="text" id="email"></td>
19 </tr>
20 <tr>
21 <td><label for="phn"><i>Phone Number</i></label></td>
22 <td><input type="text" id="phn"></td>
23 </tr>
24 <tr>
25 <td><label for="pass"><i>Password :</i></label></td>
26 <td><input type="password" id="pass"></td>
27 </tr>
28 <tr>
29 <td colspan="2" style="text-align: center;"><button Register</button> <input type="reset"></td>
30 </tr>
31 </table>
32 </div>
33 </body>
34 </html>

```

Final page:

```

1 <!DOCTYPE html>
2 <html lang="en">
3 <head>
4 <title>I FRAMES</title>
5 </head>
6 <body style="background-image: url(4b.png); margin: 80px auto; background-repeat: no-repeat; background-size: cover;">
7 <div style="text-align: center; color: black;">COLLAGE REGISTRATION FORM</div>
8 <div style="display: flex; justify-content: space-around; align-items: center;">
9 <div style="border: 1px solid black; padding: 10px; width: 40%; text-align: center;">
10 <h3>COLLAGE REGISTRATION FORM</h3>
11 </div>
12 <div style="border: 1px solid black; padding: 10px; width: 40%; text-align: center;">
13 <h3>COLLAGE REGISTRATION FORM</h3>
14 </div>
15 </div>
16 <div style="text-align: center; color: black;">
17 <div style="display: flex; justify-content: space-around; align-items: center;">
18 <div style="border: 1px solid black; padding: 10px; width: 40%; text-align: center;">
19 <h3>COLLAGE REGISTRATION FORM</h3>
20 </div>
21 <div style="border: 1px solid black; padding: 10px; width: 40%; text-align: center;">
22 <h3>COLLAGE REGISTRATION FORM</h3>
23 </div>
24 </div>
25 </div>
26 </body>
27 </html>

```

Output:



Tags used:

- <div>: this tag is used to create a section or divides specific part in webpage.
- <iframe>: this tag is used to provide separate webpages in same webpage.
- Width: this is attribute used to set the width of the iframe.
- Height: attribute used to set the height of the iframe.
- Display flex: attribute used for displaying webpages side by side.
- Back-drop filter: blur(): applies blur effect to the background.

WEEK-5

5a. To create a responsive card layout using HTML and CSS with three cards, each containing an image, title, and button.

Aim: To create a responsive card layout using HTML and CSS with three cards, each containing an image, title, and button

Program:

```

1 <!DOCTYPE html>
2 <html>
3 <head>
4 <title>Document</title>
5 <style>
6 <!--AV.sc.u4csa25187-->
7
8 .container
9 {
10     background-color: #rgb(255, 255, 255);
11     justify-content: center;
12     display: flex;
13     gap: 30px;
14     flex-wrap: wrap;
15 }
16
17 .card
18 {background-color: transparent;
19
20 padding: 20px;
21 width: 400px;
22 height: 200px;
23 border-radius: 10px;
24 box-shadow: 0 4px 10px #rgba(0,0,0,0.2);
25 text-align: center;
26 }
27
28 .card h1
29 {
30     text-align: center;
31     font-family: 'Poppins', sans-serif;
32 }
33
34 .card p
35 {
36     font-family: 'Open Sans', sans-serif;
37     text-align: center;
38     font-size: larger;
39 }
40
41 .card button
42 {
43     background-color: #rgb(164, 214, 255);
44     color: #black;
45     font-family: 'Franklin Gothic Medium', 'Arial Narrow', Arial, sans-serif;
46 }
47
48 .card button:hover
49 {
50     background-color: #1B77F2;
51 }
52
53 </style>
54 </head>
55 <body>
56 <div class="container">
57 <div class="card">

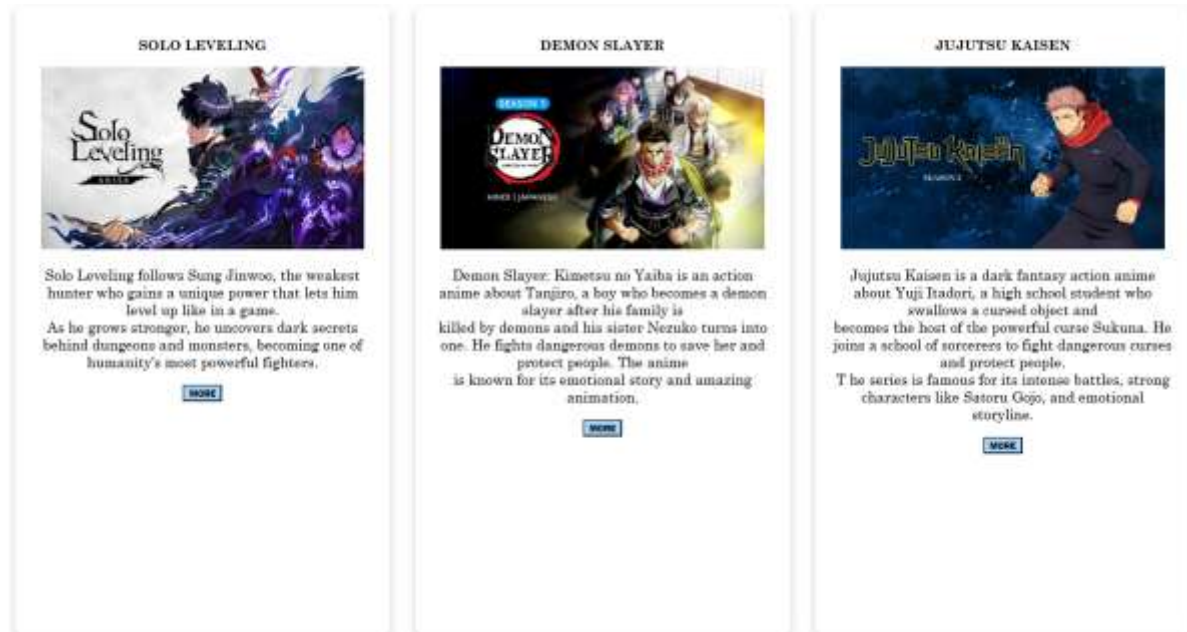
```

```

1 <html>
2 <head>
3 <body>
4   <div class="container">
5     <div class="card">
6       <h3>SOLO LEVELING</h3>
7       
8       <p>Solo leveling follows Sung Jinwoo, the weakest hunter who gains a unique power that lets him level up like in a game.</p>
9       <p>As he grows stronger, he uncovers dark secrets behind dungeons and monsters, becoming one of humanity's most powerful fighters.</p>
10      <form action="https://en.wikipedia.org/wiki/Solo_Leveling">
11        <button>MORE</button>
12      </form>
13    </div>
14
15    <div class="card">
16      <h3>DEMON SLAYER</h3>
17      
18      <p>Demon Slayer: Kimetsu no Yaiba is an action anime about Tanjiro, a boy who becomes a demon slayer after his family is</p>
19      <p>killed by demons and his sister Nezuko turns into one. He fights dangerous demons to save her and protect people. The anime is</p>
20      <p>known for its emotional story and amazing animation.</p>
21      <form action="https://en.wikipedia.org/wiki/Demon_Slayer:_Kimetsu_no_Yaiba">
22        <button>MORE</button>
23      </form>
24    </div>
25
26    <div class="card">
27      <h3>JUZUTSU KAISEN</h3>
28      
29      <p>Jujutsu Kaisen is a dark fantasy action anime about Yuji Itadori, a high school student who swallows a cursed object and</p>
30      <p>becomes the host of the powerful curse Sukuna. He joins a school of sorcerers to fight dangerous curses and protect people.</p>
31      <p>The series is famous for its intense battles, strong characters like Satoru Gojo, and emotional storylines.</p>
32      <form action="https://en.wikipedia.org/wiki/Jujutsu_Kaisen">
33        <button>MORE</button>
34      </form>
35    </div>
36  </div>
37
38 </body>
39 </html>

```


OUTPUT:



Tags Used:

- `<style>` – Used to add internal CSS styles to design the webpage.
- `<div>` – A container used to group and organize HTML elements.
- `<img>` – Displays an image on the webpage using the source URL.
- `<br>` – Inserts a line break within text.
- `<form>` – Creates a form used to send data or redirect to another page.
- `<button>` – Creates a clickable button for user interaction.
- `Hover:` is used to add the effect of glow to the button.

5b. To design a mini website using iframe with navigation links (Home, About, Contact), where clicking on each link displays the respective content within the same page using internal or external CSS.

Aim:

To design a mini website using iframe with navigation links (Home, About, Contact), where clicking on each link displays the respective content within the same page using internal or external CSS.

Program:

Main:

```

index.html X 5ch.html 5ch.css 5ch.css.html 5ch.css.html 5ch.css.html 5ch.css.html 5ch.css.html 5ch.css.html
1 <!DOCTYPE html>
2 <html>
3 <head>
4 <title></title>
5 <style>
6 {
7   text-decoration: none;
8   color: #E0E0E0;
9 }
10 :visited{
11   color: #E0E0E0;
12 }
13 </style>
14 </head>
15 <body style="background-color:rgb(33, 33, 33); color: #E0E0E0;">
16 <div style="text-align: center;">Rohit Sharma</div>
17 <div>
18 <table border="1" style="width: 100%; text-align: center;">
19 <tr>
20 <td><a href="home.html" target="box">HOME</a></td>
21 <td><a href="about.html" target="box">ABOUT</a></td>
22 <td><a href="contact.html" target="box">CONTACT</a></td>
23 </tr>
24 </table></div>
25 <div>
26 <iframe name="box" height="300" width="100%" style="background-color: #aliceblue;"></div>
27
28 </body>
29 </html>

```

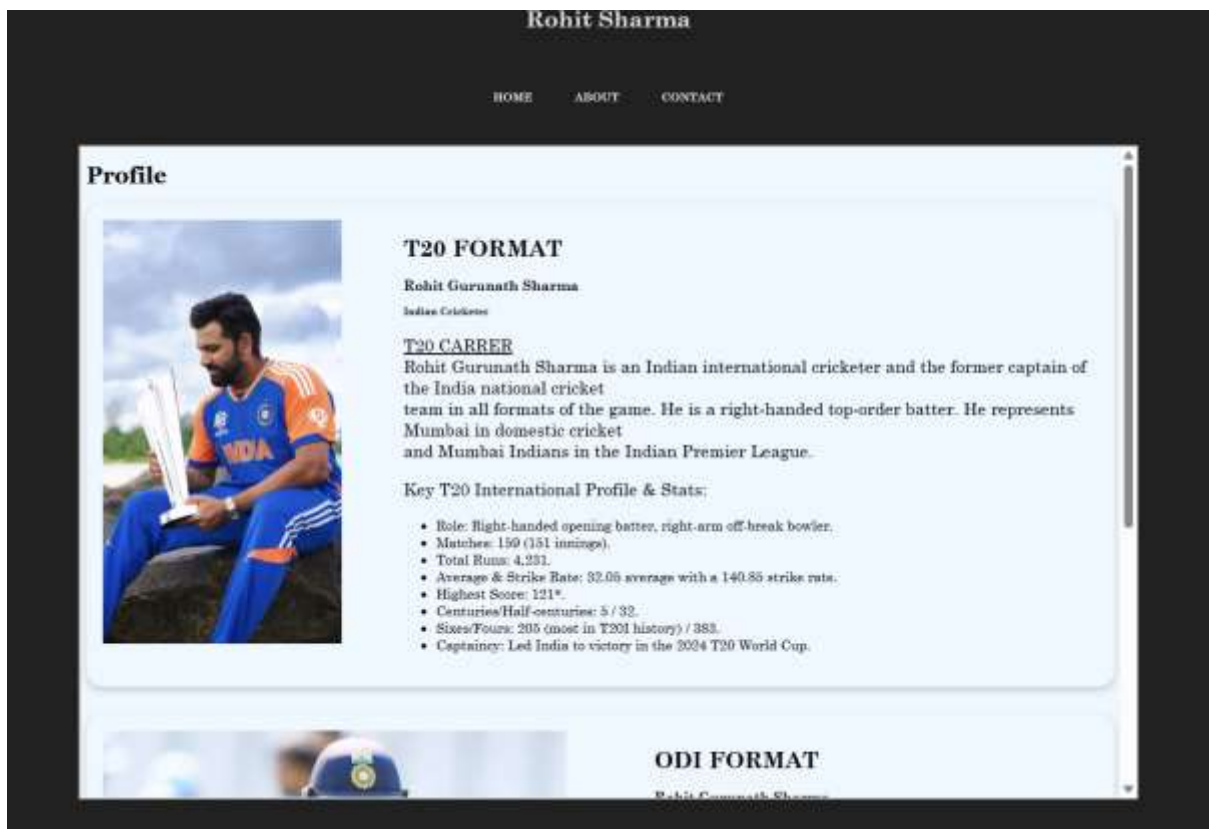
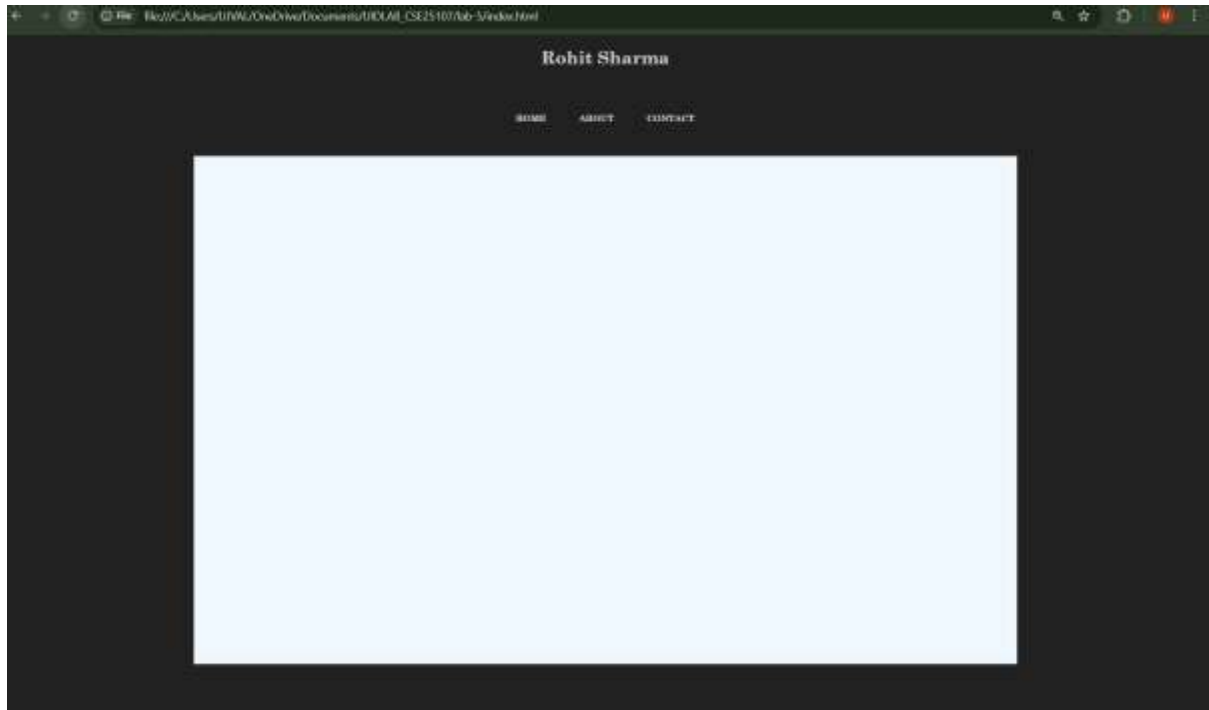
Sub webpages:

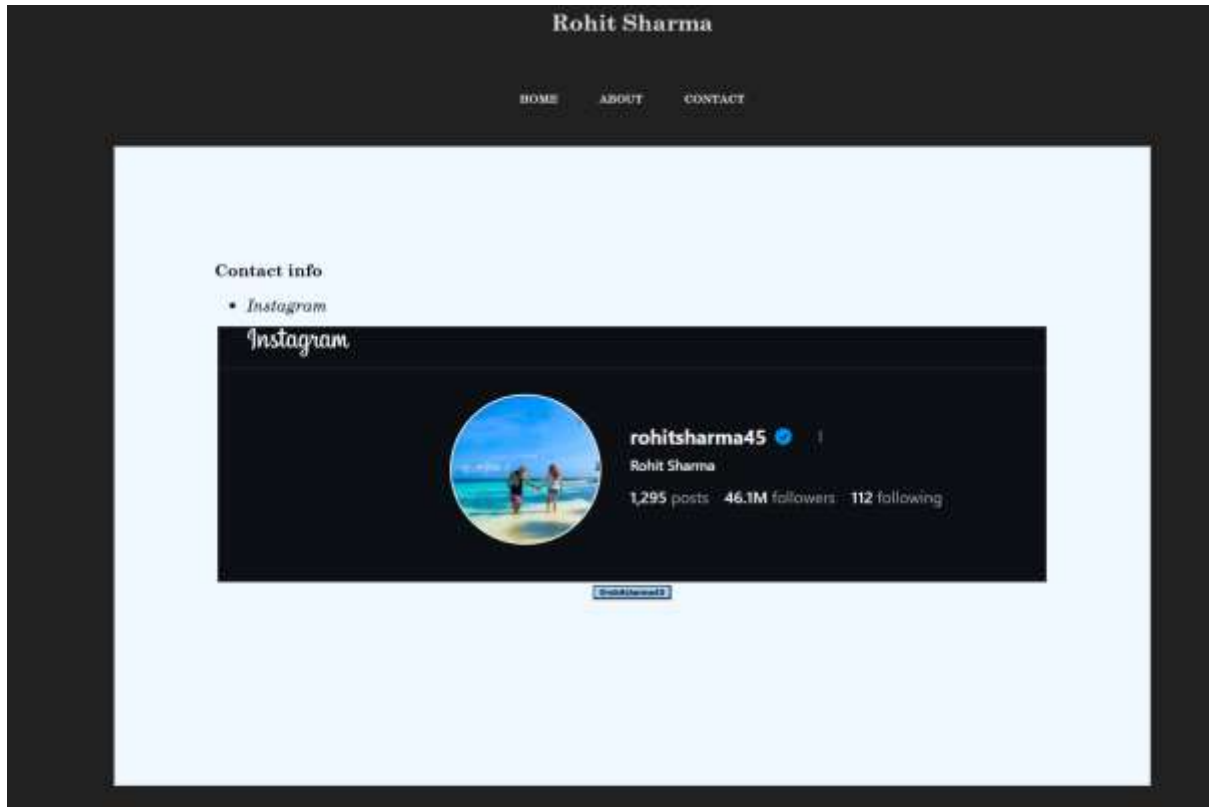
Contact:

```

1 <!DOCTYPE html>
2 <html>
3 <head>
4 <title></title>
5 <style>
6 {
7   background-color: #f0f0f0;
8   color: #000000;
9   font-family: 'Franklin Gothic Medium', 'Arial Narrow', Arial, sans-serif;
10 }
11 :hover{
12   background-color: #f0f0f0;
13 }
14 </style>
15 </head>
16 <body style="padding: 50px; text-align: center;">
17 <div>
18 <div>
19 <div style="font-size: 24px; font-weight: bold;">CONTACT</div>
20 <div>
21 <div style="display: inline-block; width: 100px; height: 100px; background-color: #f0f0f0; border: 1px solid #000; border-radius: 50%; text-align: center; vertical-align: middle;">
22 <img alt="Profile picture of Rohit Sharma" style="width: 80px; height: 80px; border-radius: 50%; border: 1px solid #000; margin: 0 auto;"/>
23 <div style="display: inline-block; vertical-align: middle; margin-left: 10px;">
24 <div style="font-weight: bold; margin: 0;">Rohit Sharma</div>
25 <div style="font-size: 12px; margin: 0;">>@rohitsharma45</div>
26 </div>
27 </div>
28 </div>
29 </div>
30 </body>
31 </html>

```



Tags used:

- <div> – A container used to group and structure HTML elements.
- – Displays an image on the webpage using a source URL.
-
 – Inserts a line break in the text.
- – Creates an unordered list of items.
- – Represents an item inside a list.
- <center> – Centers the content horizontally on the webpage.
- <table> – Creates a table structure to organize data in rows and columns.
- <iframe> – Embeds another webpage inside the current page.
- <form> – Creates a form used for submitting data or redirecting to another page.
- <button> – Creates a clickable button for user interaction.
- Hover: is used to add the effect of glow to the button.

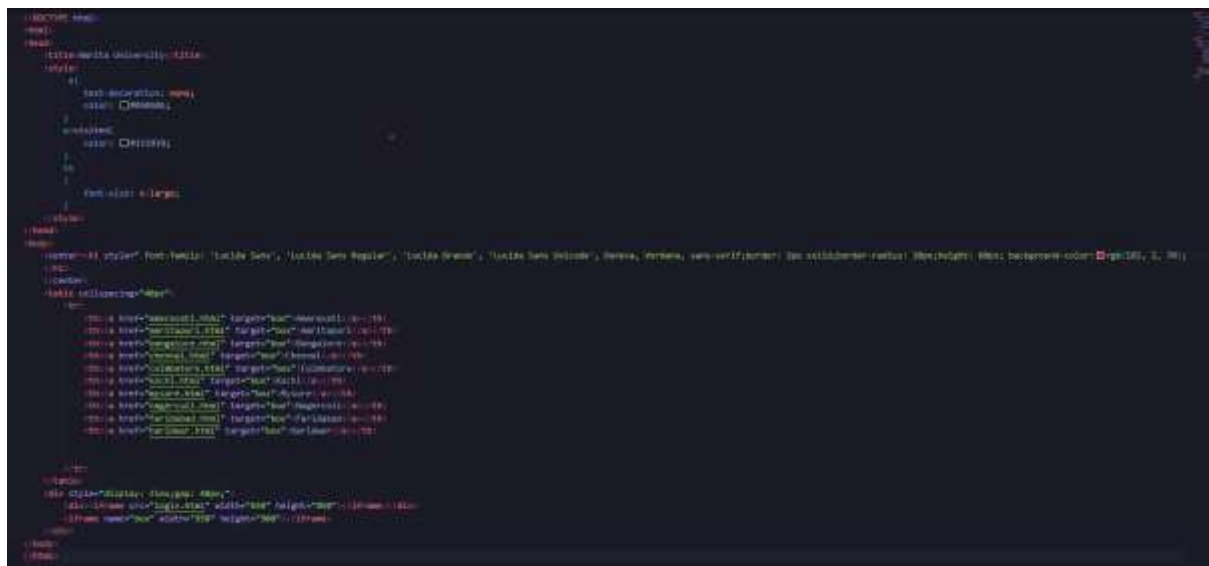
5c. To design an Amrita login page using HTML and CSS that displays different campuses, where clicking on each campus shows its details on the same page.

Aim:

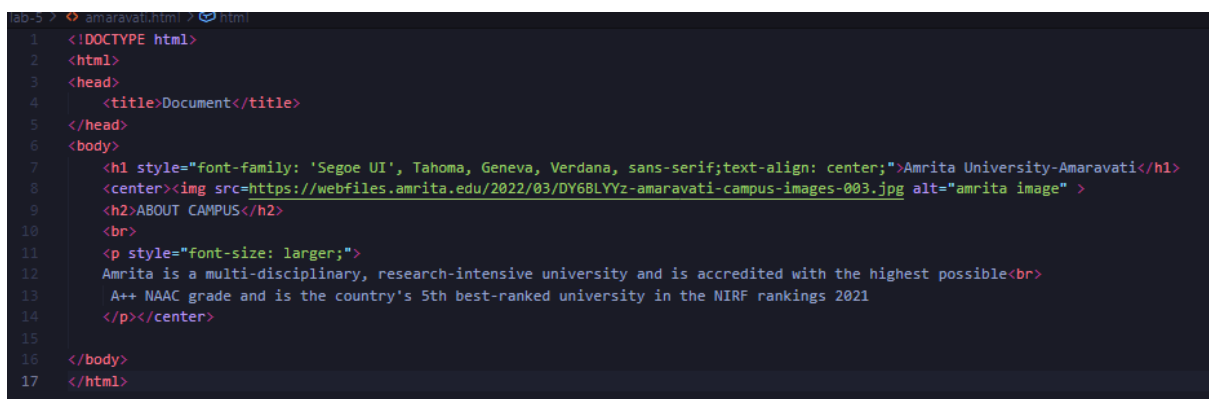
To design an Amrita login page using HTML and CSS that displays different campuses, where clicking on each campus shows its details on the same page.

Program:

Main:



Sub webpages:



```

1 <!DOCTYPE html>
2 <html>
3 <head>
4   <title>Document</title>
5 </head>
6 <body>
7   <h1 style="font-family: 'Segoe UI', Tahoma, Geneva, Verdana, sans-serif;text-align: center;">Amrita University-Amritapuri</h1>
8   <center><img src=https://webfiles.amrita.edu/2022/02/f5229Wqz-unesco-banner-01.jpg alt="amrita image" >
9   <h2>ABOUT CAMPUS</h2>
10  <br>
11  <p style="font-size: larger;">
12    Amrita is a multi-disciplinary, research-intensive university and is accredited with the highest possible<br>
13    A++ NAAC grade and is the country's 5th best-ranked university in the NIRF rankings 2021<br>
14    Amritapuri (Malayalam: അമൃതപുരി, Sanskrit: अमृतपुरी), originally Parayakadavu, is the main ashram of Indian<br>
15    Hindu spiritual leader, guru and humanitarian Mata Amritanandamayi Devi, often known as Amma ("Mother"), who<br>
16    is revered as 'the hugging saint' by her followers.
17  </p></center>
18 </body>
19 </html>

```

```

1 <!DOCTYPE html>
2 <html>
3 <head>
4   <title>Document</title>
5 </head>
6 <body>
7   <h1 style="font-family: 'Segoe UI', Tahoma, Geneva, Verdana, sans-serif;text-align: center;">Amrita University-Bangalore</h1>
8   <center><img src=https://webfiles.amrita.edu/2019/11/qCWEOMjs-amrita_bangaluru_banner.jpg alt="amrita image" >
9   <h2>ABOUT CAMPUS</h2>
10  <br>
11  <p style="font-size: larger;">
12    Amrita is a multi-disciplinary, research-intensive university and is accredited with the highest possible<br>
13    A++ NAAC grade and is the country's 5th best-ranked university in the NIRF rankings 2021<br>
14    achieving 8th rank among Indian Universities in NIRF 2025 and 7th in NIRF 2024
15  </p></center>
16 </body>
17 </html>

```

```

1 <!DOCTYPE html>
2 <html>
3 <head>
4   <title>Document</title>
5 </head>
6 <body>
7   <h1 style="font-family: 'Segoe UI', Tahoma, Geneva, Verdana, sans-serif;text-align: center;">Amrita University-CHENNAI</h1>
8   <center><img src=https://webfiles.amrita.edu/2019/06/VVENH34U-amrita-vishwa-vidyapeetham-chennai-campus-banner.jpg alt="amrita image" >
9   <h2>ABOUT CAMPUS</h2>
10  <br>
11  <p style="font-size: larger;">
12    Amrita is a multi-disciplinary, research-intensive university and is accredited with the highest possible<br>
13    A++ NAAC grade and is the country's 5th best-ranked university in the NIRF rankings 2021<br>
14    achieving 8th rank among Indian Universities in NIRF 2025 and 7th in NIRF 2024
15  </p></center>
16 </body>
17 </html>

```

```

1 <!DOCTYPE html>
2 <html>
3 <head>
4   <title>Document</title>
5 </head>
6 <body>
7   <h1 style="font-family: 'Segoe UI', Tahoma, Geneva, Verdana, sans-serif;text-align: center;">Amrita University-Coimbatore</h1>
8   <center><img src=https://webfiles.amrita.edu/2020/06/nyYIkquw-790-100.jpg alt="amrita image" >
9   <h2>ABOUT CAMPUS</h2>
10  <br>
11  <p style="font-size: larger;">
12    Amrita is a multi-disciplinary, research-intensive university and is accredited with the highest possible<br>
13    A++ NAAC grade and is the country's 5th best-ranked university in the NIRF rankings 2021<br>
14    achieving 8th rank among Indian Universities in NIRF 2025 and 7th in NIRF 2024
15  </p></center>
16 </body>
17 </html>

```

```
1 <!DOCTYPE html>
2 <html>
3 <head>
4   <title>Document</title>
5 </head>
6 <body>
7   <h1 style="font-family: 'Segoe UI', Tahoma, Geneva, Verdana, sans-serif;text-align: center;">Amrita University-Faridabad</h1>
8   <center>
9   </center><h2>ABOUT CAMPUS</h2>
10   <div>
11     <p style="font-size: larger;">
12       Amrita is a multi-disciplinary, research-intensive university and is accredited with the highest possible <br>
13       A++ NAAC grade and is the country's 5th best-ranked university in the NIRF rankings 2021 <br>
14       achieving 8th rank among Indian Universities in NIRF 2025 and 7th in NIRF 2024
15     </p></center>
16   </div>
17 </body>
18 </html>
```

```

1 <!DOCTYPE html>
2 <html>
3 <head>
4   <title>Document</title>
5 </head>
6 <body>
7   <h1 style="font-family: 'Segoe UI', Tahoma, Geneva, Verdana, sans-serif;text-align: center;">Amrita University-Haridwar</h1>
8   <center>
9   </center><h2>ABOUT CAMPUS</h2>
10  <br>
11  <p style="font-size: larger;">
12    Amrita is a multi-disciplinary, research-intensive university and is accredited with the highest possible<br>
13    A++ NAAC grade and is the country's 5th best-ranked university in the NIRF rankings 2021<br>
14    achieving 8th rank among Indian Universities in NIRF 2025 and 7th in NIRF 2024
15  </p></center>
16
17 </body>
18 </html>

```

[illegible]

Output:

Amrita university

[Amaravati](#)
[Amritapuri](#)
[Bangalore](#)
[Chennai](#)
[Coimbatore](#)
[Kochi](#)
[Mysore](#)
[Nagercoil](#)
[Faridabad](#)
[Haridwar](#)

COLLEGE ADMISSION REGISTRATION	
Full Name:	
Email:	
Phone Number:	
Gender:	☐ Male ☐ Female ☐ other
Course To Apply:	
Subject:	☐ MEDICAL ☐ SPORTS ☐ READING ☐
Date Of Birth:	
Upload:	
Address:	
Submit Reset	

Amrita university

[Amaravati](#)
[Amritapuri](#)
[Bangalore](#)
[Chennai](#)
[Coimbatore](#)
[Kochi](#)
[Mysore](#)
[Nagercoil](#)
[Faridabad](#)
[Haridwar](#)

COLLEGE ADMISSION REGISTRATION	
Full Name:	
Email:	
Phone Number:	
Gender:	☐ Male ☐ Female ☐ other
Course To Apply:	
Subject:	☐ MEDICAL ☐ SPORTS ☐ READING ☐
Date Of Birth:	
Upload:	
Address:	
Submit Reset	

Amrita University-Kochi



ABOUT CAMPUS

Amrita is a multi-disciplinary, research-intensive university and is accredited with the highest possible A++ NAAC grade and is the country's top best ranked university in the NIRF rankings 2021 achieving 9th rank among Indian Universities in NIRF 2020 and 7th in NIRF 2024.

Tags-used:

- `<div>` – A container used to group and structure HTML elements.
- `<img>` – Displays an image on the webpage using a source URL.
- `<br>` – Inserts a line break in the text.
- `<ul>` – Creates an unordered list of items.
- `<li>` – Represents an item inside a list.
- `<center>` – Centers the content horizontally on the webpage.
- `<table>` – Creates a table structure to organize data in rows and columns.
- `<iframe>` – Embeds another webpage inside the current page.
- `<form>` – Creates a form used for submitting data or redirecting to another page.
- `<button>` – Creates a clickable button for user interaction.
- Hover: is used to add the effect of glow to the button.

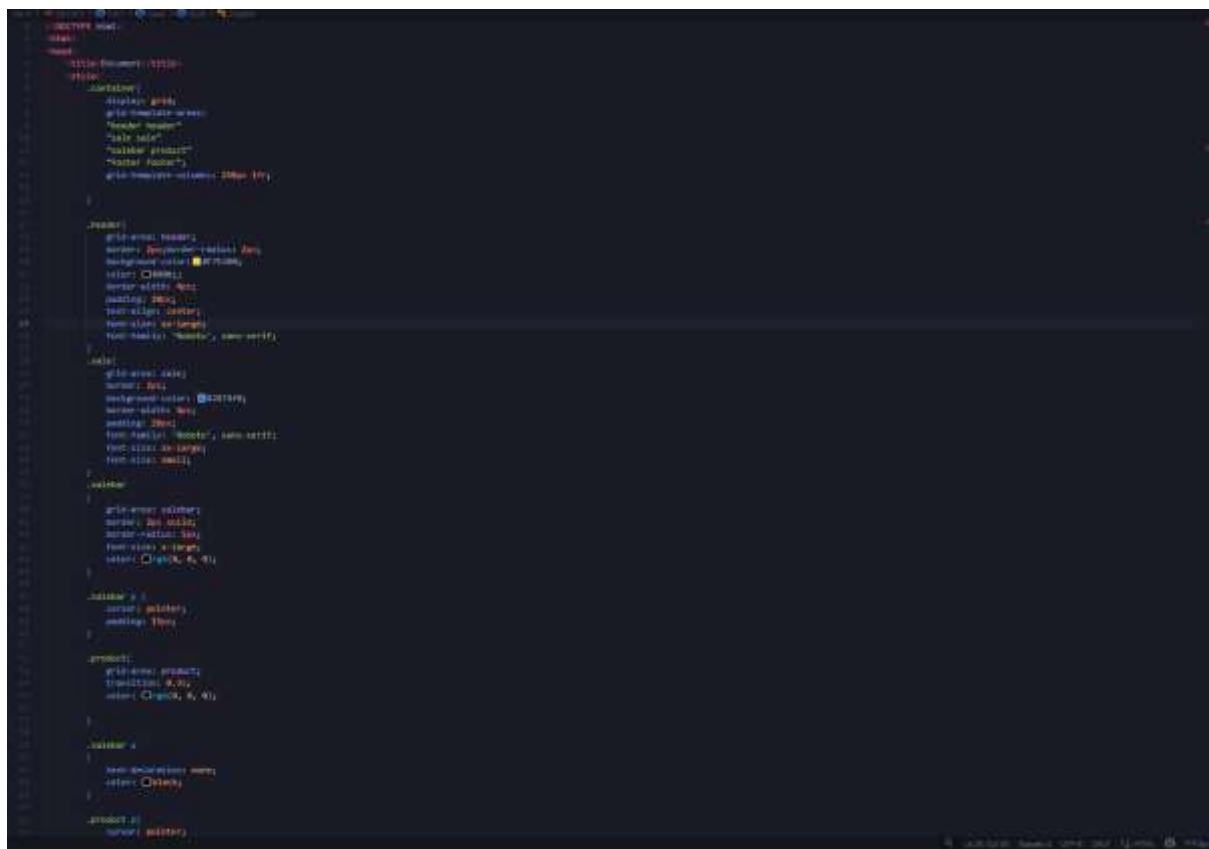
Week-6:

6. Design a responsive e-commerce application using the grid layout. This application should contain different categories of products such as electronics, clothing, home-appliances, books. Based on the category we select corresponding products should be displayed on the sidebar.

Aim:

Design a responsive e-commerce application using the grid layout. This application should contain different categories of products such as electronics, clothing, home-appliances, books. Based on the category we select corresponding products should be displayed on the sidebar.

Program:



```

<head>
<head>
<style>

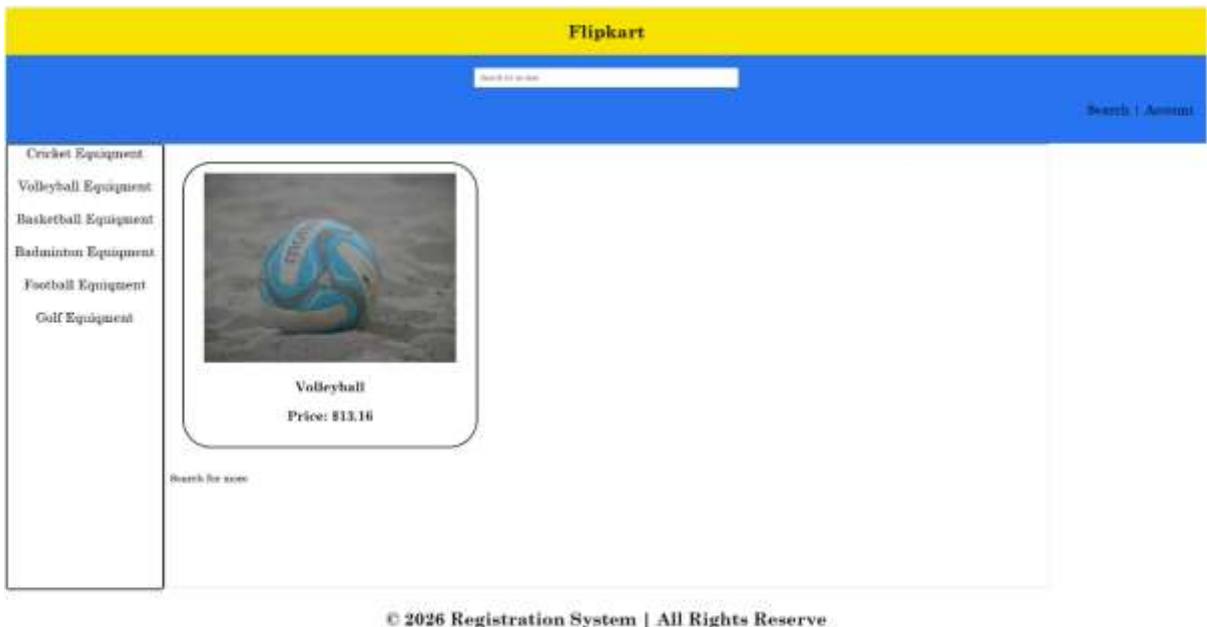
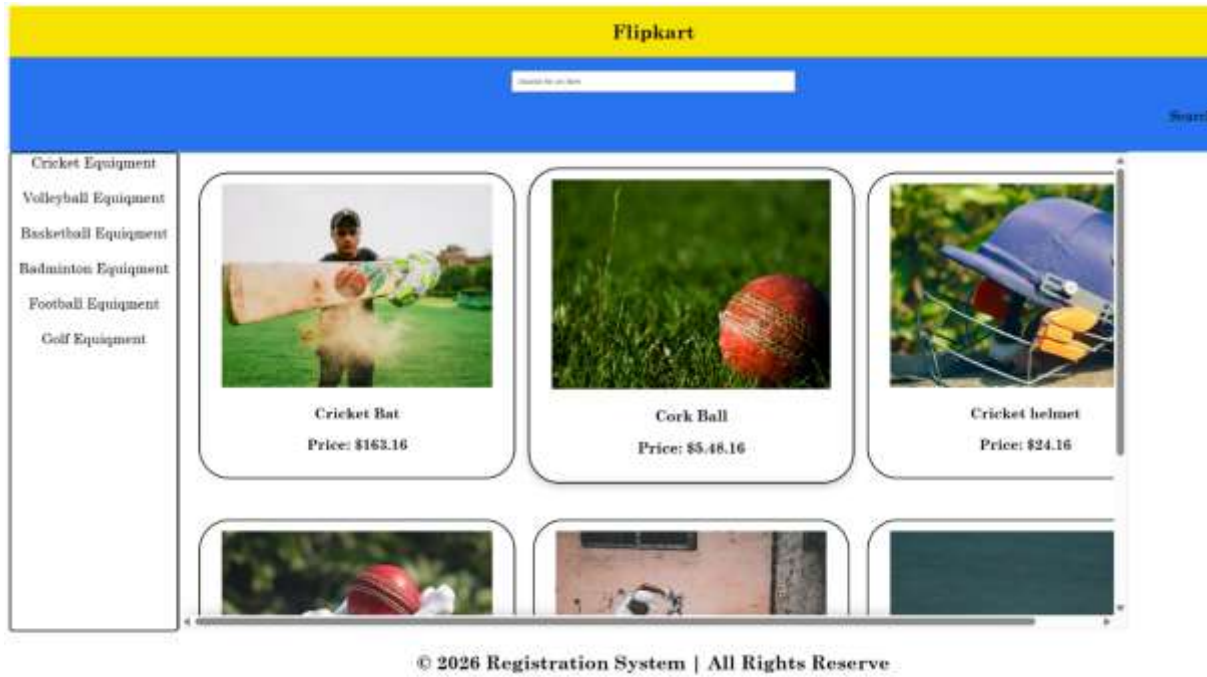
.footer{
    grid-area: footer;
    font-size: 16px;
    color: #4f4f4f;
    text-align: center;
}

.item{
    border: 2px solid;
    border-radius: 40px;
    width: 100%;
    text-align: center;
    font-size: 24px;
}

</style>
<!--
<body style="background-color: #e0e0e0;">
<div class="container">
<div class="header">
<div class="logo">
<div class="nav">
<div class="search">
<div class="button">
<div class="product">
<div class="footer">

```

Output:



Tags-used:

- **<style>** – Used to define internal CSS styles for the webpage layout and design.
- **<div>** – A container element used to group and structure sections of the webpage.
- **** – Displays text in bold format to emphasize important text.
- **<form>** – Creates a form used to collect input or send data to a server or URL.
- **<input>** – Creates an input field where users can enter data (like search text).
- **<center>** – Aligns the content horizontally at the center of the page.
- **<iframe>** – Embeds another webpage or document inside the current webpage

WEEK-7

7. Create a Git hub account and create a repository named UIDLAB upload every week experiments , lab reports in that repository and make the repository

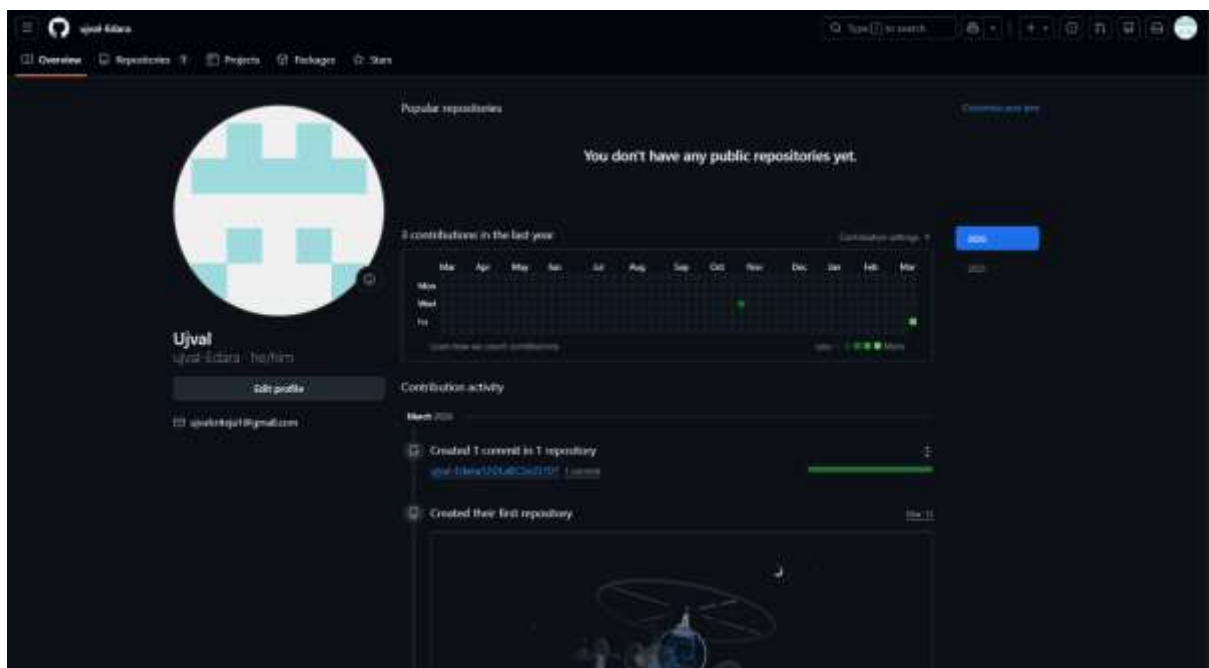
Aim: Saving codes in GitHub

Step-1:

Visit the GITHUB website <https://github.com/> .

Step-2:

Create a account or login a GIT HUB account.



Step-3:

Add repository with name which consist of UID lab and roll number.

Keep the repository to private.

Home repository

Create a new repository

Repositories contain a project's files and version history from a project's checkout. [Request a repository](#).
Required fields are marked with an asterisk (*).

General

Owner *  Repository name * UIDLaBCSe25107
UIDLaBCSe25107 is available

Great repository names are short and memorable. How does a [typo test](#) sound?

Description

1-255 characters

Configuration

Choose visibility * Private
1 contributor can see and commit to this repository

Add README READMEs can be used as longer descriptions. [About READMEs](#) Off

Add .gitignore .gitignore tells git which files not to track. [About .gitignore files](#) No .gitignore

Add license Licenses explain how others can use your code. [About licenses](#) No license

Create repository

Home repository

UIDLaBCSe25107

Code Issues Pull requests Actions Projects Security Insights Settings

main 1 branch 0 tags

UIDLaBCSe25107

1 commit 11 hours ago

README.md 1 commit 12 hours ago

README

UIDLaBCSe25107

About

No description, website, or topics provided.

Readme

Activity

0 stars

0 watching

0 forks

Releases

No releases provided.

Check for updates

Packages

No packages published.

Submit your first package

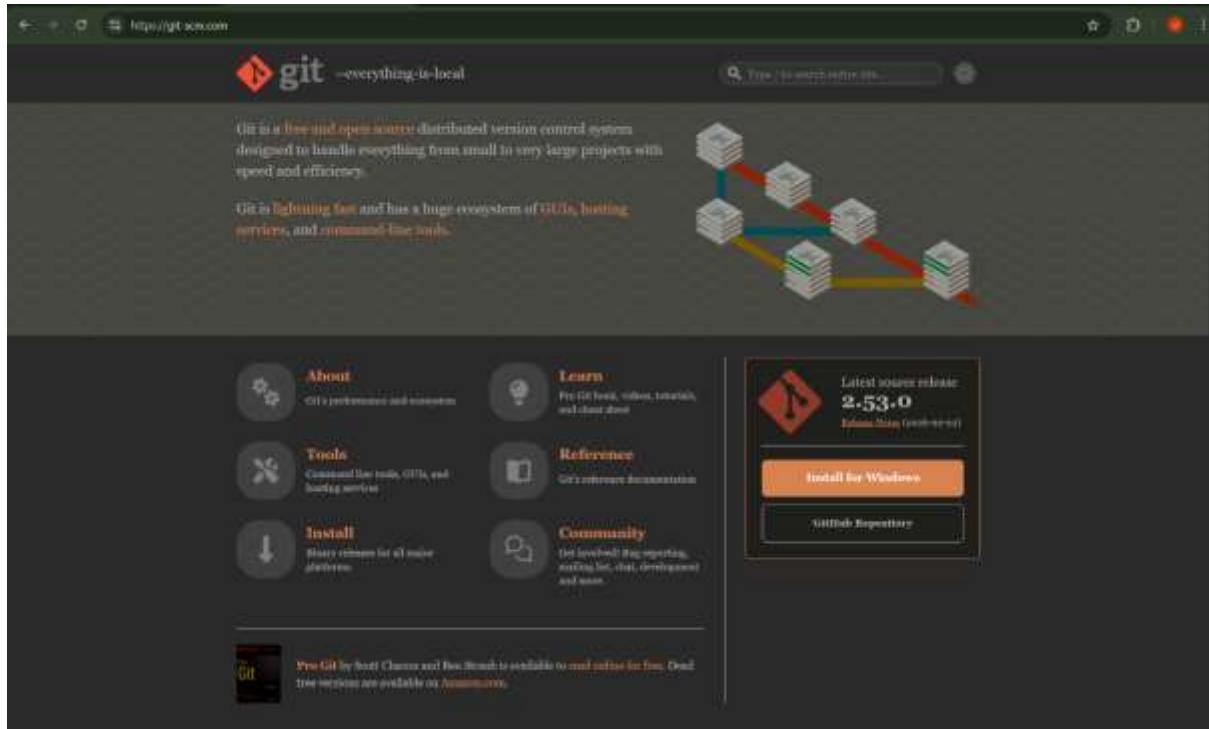
Contributors

iqual-idea

STEP-4:

Download Git Application from <https://git-scm.com/install/windows>

Restart the system after installation.

**STEP-5:**

Open in VS code the UID lab folder required to push to GitHub.

Open new Terminal In VS code.

STEP-7:

Pass these codes in the terminal to push the files to GitHub.

`git init`

`git add .`

`git commit -m "First commit"`

`git branch -M main`

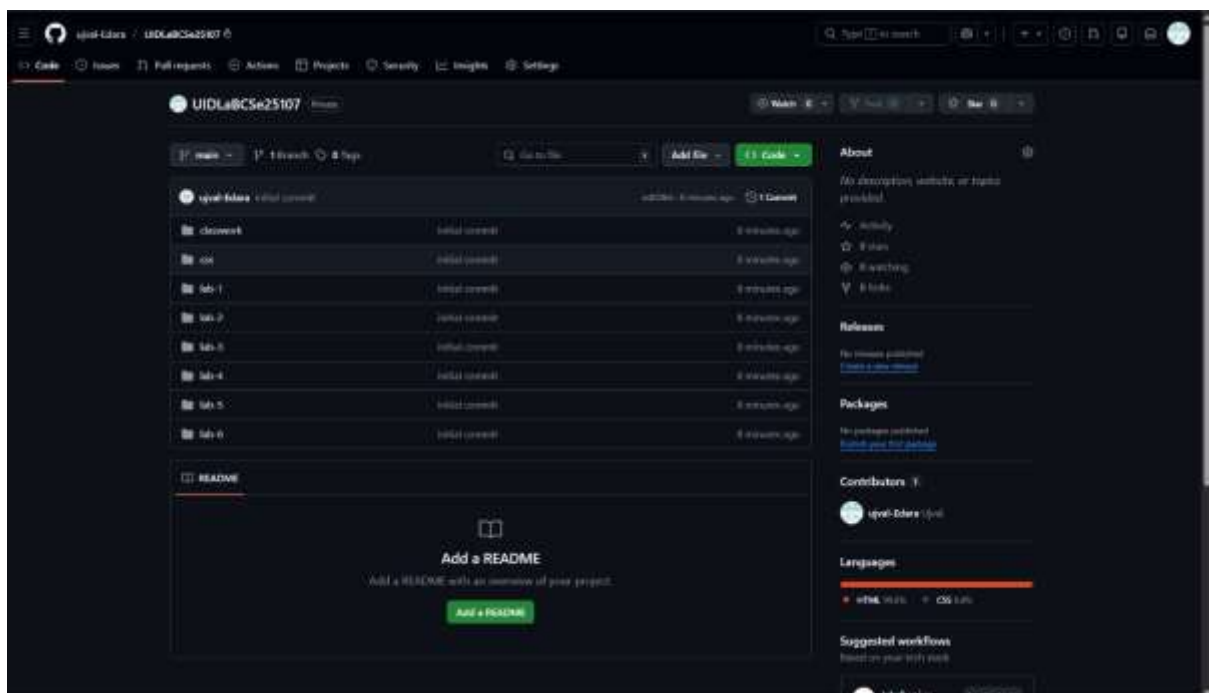
`git remote add origin <PASTE_YOUR_URL_HERE>`

`git push -u origin main`



STEP-8:

All your files will be uploaded to github.



WEEK-8

8A:

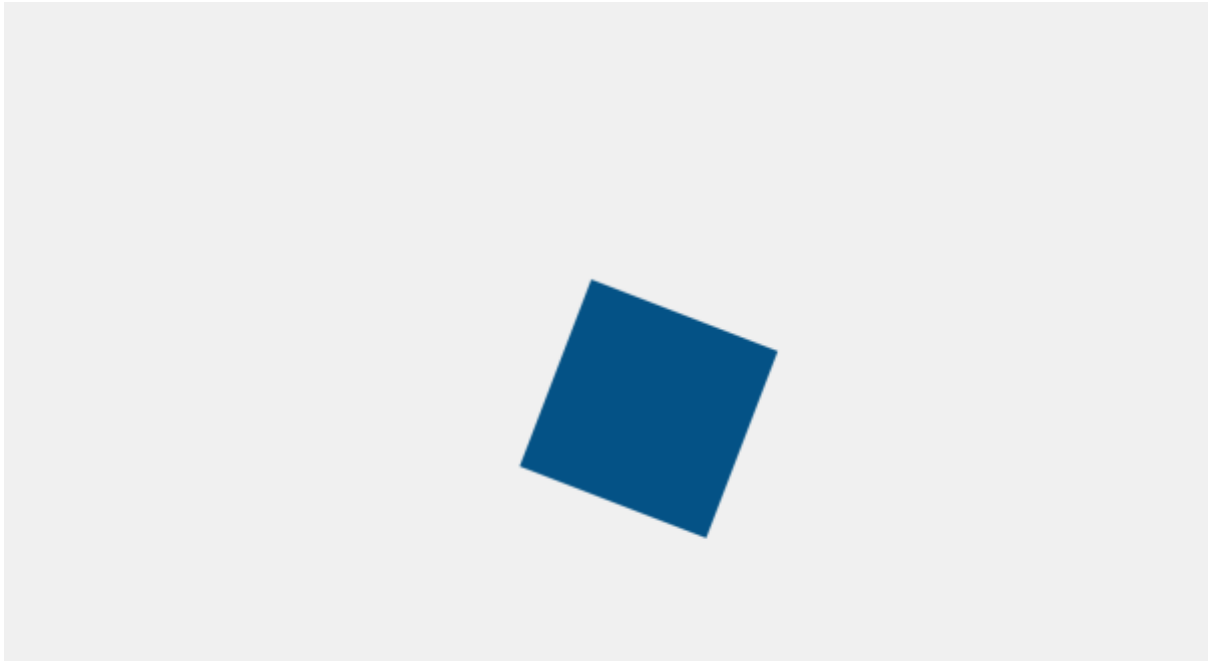
Create a webpage that display a square continuously rotate 360deg using css animations.

AIM:

Create a webpage that display a square continuously rotate 360deg using css animations .

Program:

```
lab-8 > animation.html > html > head > style
3  <html>
4
5  <head>
6  <title>Rotating Square</title>
7
8  <style>
9
10 body {
11
12 display: flex;
13
14 justify-content: center;
15 align-items: center;
16
17 height: 100vh;
18 background-color: #f0f0f0;
19
20 }
21
22
23
24 .square {
25
26 width: 200px;
27
28 height: 200px;
29
30 background-color: #045286;
31
32 animation: rotateSquare 2s linear infinite;
33 }
34
35
36 @keyframes rotateSquare {
37
38 0% {
39
40 transform: rotate(0deg);
41 }
42
43 50% {
44
45 transform: rotate(180deg);
46 }
47 100%{
48 transform: rotate(90deg);
49 }
50
51 }
52
53 </style>
54
55 </head>
56
57 <body>
58
59
60
61 <div class="square"></div>
62
63
64
65 </body>
66
67 </html>
68
```

OUTPUT:Explanation:

- `display: flex` - enables flexible layout.
- `justify-content: center` - centers horizontally.
- `align-items: center` - centers vertically.
- `height: 100vh` - full screen height.
- `background-color` - sets background color.
- Width and height used for square size.
- `background-color` - sets square color .
- `animation` - applies rotation animation.
- `rotateSquare` - animation name
- `2s` - duration (2 seconds)
- `linear` - smooth constant speed
- `infinite` - repeats forever
- `@keyframes` defines the animation steps
- `transform: rotate()` is used to rotate the square

8b: create a ball that moves up and down continuously using css animations.

AIM: create a ball that moves up and down continuously using css animations.

PROGRAM:

[illegible]

OUTPUT:



EXPLANATION:

- display: flex - enables flexible layout.
- justify-content: center - centers horizontally.
- align-items: center - centers vertically.
- height: 800px - sets page height.
- background-color - sets white background.
- width and height - define ball size.
- background-color - sets ball color.
- border-radius: 50% - makes the shape circular.
- animation - applies bouncing effect.
- bounce - animation name.
- 1s - duration (1 second).
- ease-in-out - smooth start and end.
- infinite - repeats continuously.
- @keyframes defines animation steps.
- transform: translateY() moves the ball vertically.

8c: _create a circle that moves horizontal and rotate at the same time.

AIM:

create a circle that moves horizontal and rotate at the same time.

PROGRAM:

```

lab-8 > 8.3.HTML > html > head > style > .circle
1  <!DOCTYPE html>
2
3  <html>
4
5  <head>
6
7    <title>Moving & Rotating Circle</title>
8
9    <style>
10     body {
11
12       height: 100vh;
13
14       display: flex;
15       align-items: center;
16
17       background-color: #83ac68;
18
19       overflow: hidden;
20
21     }
22
23
24
25     .circle {
26
27       width: 80px;
28
29       height: 80px;
30
31       background-color: #ffffff;
32
33       border-radius: 50%;
34       animation: moveRotate 3s linear infinite;
35
36     }
37
38
39
40     @keyframes moveRotate {
41
42       0% {
43
44         transform: translateX(0) rotate(0deg);
45
46       }
47
48       50% {
49
50         transform: translateX(300px) rotate(180deg);
51
52       }
53       100% {
54
55         transform: translateX(0) rotate(360deg);
56
57       }
58     }
59   }
60 </style>
61
62 </head>
63
64 <body>

```

OUTPUT:EXPLANATION:

- height: 100vh - full screen height
- display: flex - enables flexible layout
- align-items: center - centers vertically
- background-color - sets background color
- overflow: hidden - prevents scrollbars during animation

WEEK-9

9A:

_Create a webpage using html, CSS, java script based on the given conditions:

- a) Create a button for each program, based on the button clicked the corresponding program has to be executed and should display the output.
- b) The logics for each button include finding an even numbers, odd number, prime or not, factorial of a number, amstrong or not.
- c)Use appropriate designs to display the output.

AIM:

Create a webpage using html, CSS, java script based on the given conditions:

- a) Create a button for each program, based on the button clicked the corresponding program has to be executed and should display the output.
- b) The logics for each button include finding an even numbers, odd number, prime or not, factorial of a number, amstrong or not.
- c)Use appropriate designs to display the output.

PROGRAM:

```

1  <!DOCTYPE html>
2  <html>
3  <head>
4      <title>NUMBERS</title>
5
6      <style>
7          body {
8              font-family: Arial;
9              background: rgb(126, 233, 219);
10             text-align: center;
11         }
12
13         .main {
14             background: rgb(94, 94, 94);
15             width: 320px;
16             margin: 80px auto;
17             padding: 25px;
18             border-radius: 12px;
19             box-shadow: 0 4px 10px rgb(246, 242, 242);
20         }
21
22         input {
23             padding: 10px;
24             width: 80%;
25             margin-bottom: 15px;
26         }
27
28         button {
29             padding: 8px 12px;
30             margin: 4px;
31             border: none;
32             border-radius: 5px;
33             background: #cbdaf9;
34             color: rgb(0, 0, 0);
35             cursor: pointer;
36         }
37
38         button:hover {
39             background: #000000;
40         }
41
42         #result {
43             margin-top: 15px;
44             font-weight: bold;
45             color: rgb(101, 245, 243);

```

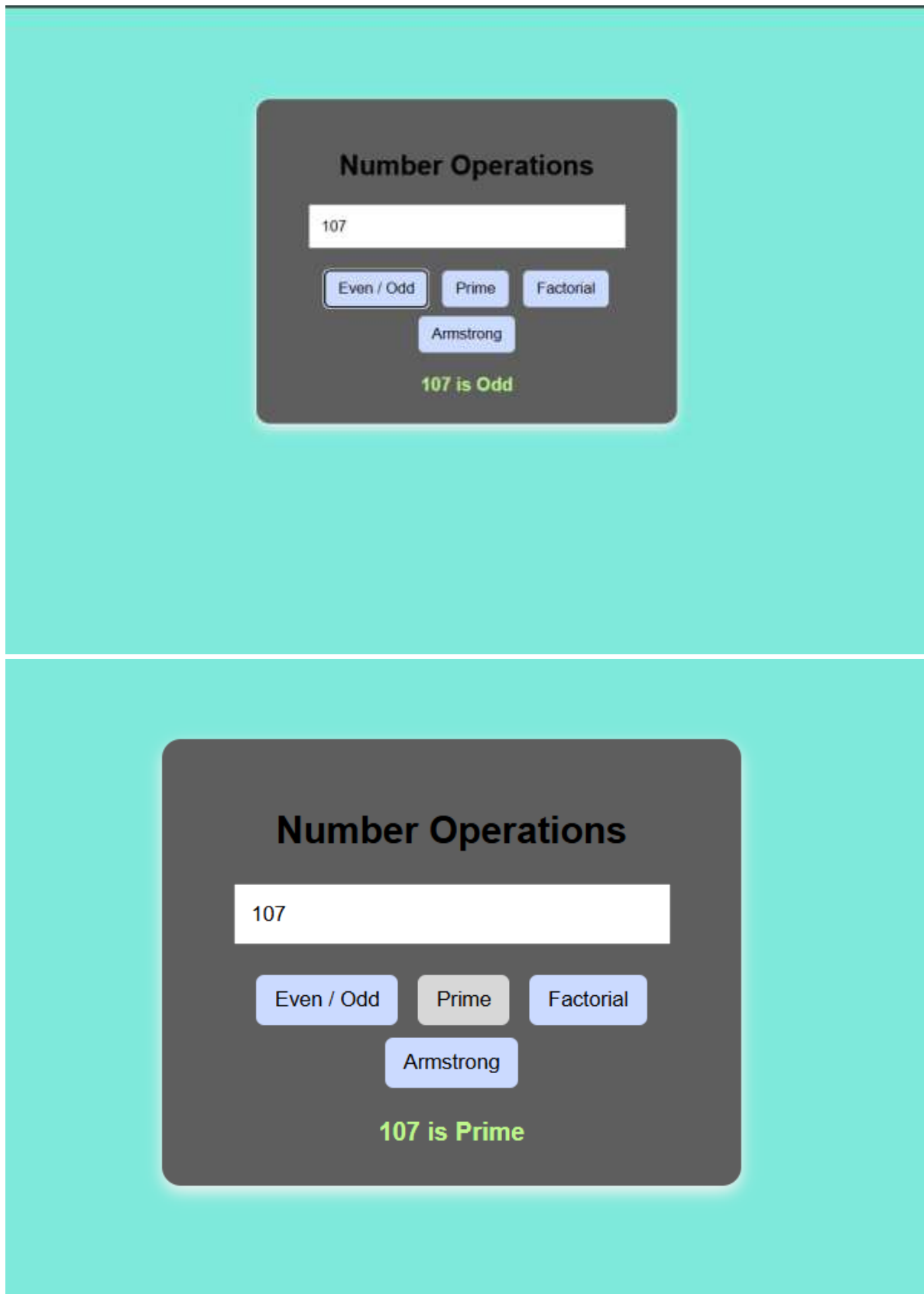
```
Lab-9 > lab9a.html > html > head > style > .main
2  <html>
3  <head>
6    <style>
42      #result {
43        margin-top: 15px;
44        font-weight: bold;
45        color: ■rgb(191, 245, 141);
46      }
47    </style>
48  </head>
49
50  <body>
51
52  <div class="main">
53    <h2>Number Operations</h2>
54
55    <input type="number" id="num" placeholder="Enter a number">
56
57    <br>
58
59    <button onclick="evenOdd()">Even / Odd</button>
60    <button onclick="primeCheck()">Prime</button>
61    <button onclick="factorial()">Factorial</button>
62    <button onclick="armstrong()">Armstrong</button>
63
64    <div id="result">THIS IS THE PLACE OF OUTPUT</div>
65  </div>
66
67  <script>
68
69
70  function getNumber() {
71    return Number(document.getElementById("num").value);
72  }
73
74  function show(msg) {
75    document.getElementById("result").innerHTML = msg;
76  }
77
78
79  function evenOdd() {
80    let n = getNumber();
81
82    if (n % 2 === 0)
```

lab9a.html > html > head > style > .main

```
2    <html>
50   <body>
67   <script>
78
79   function evenOdd() {
80       let n = getNumber();
81
82       if (n % 2 === 0)
83           show(n + " is Even");
84       else
85           show(n + " is Odd");
86   }
87
88
89   function primeCheck() {
90       let n = getNumber();
91       let isPrime = true;
92
93       if (n <= 1) isPrime = false;
94
95       for (let i = 2; i < n; i++) {
96           if (n % i === 0) {
97               isPrime = false;
98               break;
99           }
100      }
101
102      if (isPrime)
103          show(n + " is Prime");
104      else
105          show(n + " is Not Prime");
106  }
107
108
109  function factorial() {
110      let n = getNumber();
111      let f = 1;
112
113      for (let i = 1; i <= n; i++) {
114          f *= i;
115      }
116
117      show("Factorial = " + f);
118  }
```

```
5-9 > <> lab9a.html > html > head > style > .main
2   <html>
30  <body>
37  <script>
39  function factorial() {
47  |     show("Factorial = " + t);
48  | }
49
50  function armstrong() {
51  |     let n = getNumber();
52  |     let temp = n;
53  |     let sum = 0;
54  |
55  |     while (temp > 0) {
56  |         let digit = temp % 10;
57  |         sum += digit * digit * digit;
58  |         temp = Math.floor(temp / 10);
59  |     }
60  |
61  |     if (sum === n)
62  |         show(n + " is Armstrong");
63  |     else
64  |         show(n + " is Not Armstrong");
65  | }
66
67  </script>
68
69  </body>
70  </html>
```

OUTPUT:



Number Operations

107

Even / Odd

Prime

Factorial

Armstrong

Factorial = 1.2265202031961373e+172

Number Operations

107

Even / Odd

Prime

Factorial

Armstrong

107 is Not Armstrong

Explanation:

- `<input>` is used to take number input from the user
- `<button>` is used to perform different operations
- `<div>` is used to display the output

In java script:

- `getNumber` – this is the function created to get the number entered by the user.\
- `show(msg)` – this function is created to display output on the div element.
- `evenOdd` – this function is created to verify weather the user input is even or odd and the code in the function verifies checks the number.
- `primeCheck` – this is the function created for checking the prime and composite for given user input and its check by the code in the function.
- `Factorial` – this function is to generate factorial for ser input by code present in the function.
- `Armstrong` – this function used to check weather the given number is armstong or not by the code present in function.

WEEK-10

10a: write a Java script conditional statement to find the sign of product of three numbers using function. Display an alert box with specific sign

AIM: write a Java script conditional statement to find the sign of product of three numbers using function. Display an alert box with specific sign

PROGRAM:

```
Lab-10 > 10a.html > html > body
1  <!DOCTYPE html>
2  <html lang="en">
3  <head>
4      <title>Product Sign Checker</title>
5      <link rel="stylesheet" href="style.css">
6  </head>
7  <body>
8
9      <div class="glass-card">
10         <h2>Sign Checker</h2>
11         <p>Determine the sign of the product</p>
12
13         <input type="number" id="num1" placeholder="Enter first number">
14         <input type="number" id="num2" placeholder="Enter second number">
15         <input type="number" id="num3" placeholder="Enter third number">
16
17         <button onclick="checkSign()">Check Sign</button>
18
19         <div id="result"></div>
20     </div>
21
22
23     <script>
24         function checkSign() {
25             const n1 = parseFloat(document.getElementById('num1').value);
26             const n2 = parseFloat(document.getElementById('num2').value);
27             const n3 = parseFloat(document.getElementById('num3').value);
28             const display = document.getElementById('result');
29
30             if (isNaN(n1) || isNaN(n2) || isNaN(n3)) {
31                 display.innerText = "Fill all fields";
32                 return;
33             }
34
35             const product = n1 * n2 * n3;
36
37             if (product > 0) {
38                 alert("The sign is [+]");
39             } else if (product < 0) {
40                 alert("The sign is [-]");
41             } else {
42                 alert("The product is 0");
43             }
44         }
45     </script>
46 </body>
47 </html>
```

OUTPUT:

The image shows a web application titled "Sign Checker" with the subtitle "Determine the sign of the product". It features three input fields containing the values "23", "-8", and "6". Below these fields is a blue "Check Sign" button and a "Fill all fields" link. A dark overlay at the bottom of the screen displays the message "This page says The sign is [-]" with an "OK" button.

Sign Checker
Determine the sign of the product

23

-8

6

Check Sign

Fill all fields

This page says
The sign is [-]

OK

EXPLANATION:

- Takes **3 numbers from user**
- type="number" - only numeric input allowed
- id - used in JavaScript to access values
- placeholder - hint text
- ON clicked - calls the JavaScript function checkSign().
- document.getElementById() - selects input field.
- value - gets user input.
- parseFloat() - converts string → number.

10b:

To Sort 3 number using functions. Display on alert box to show the result.

AIM:

To Sort 3 number using functions. Display on alert box to show the result.

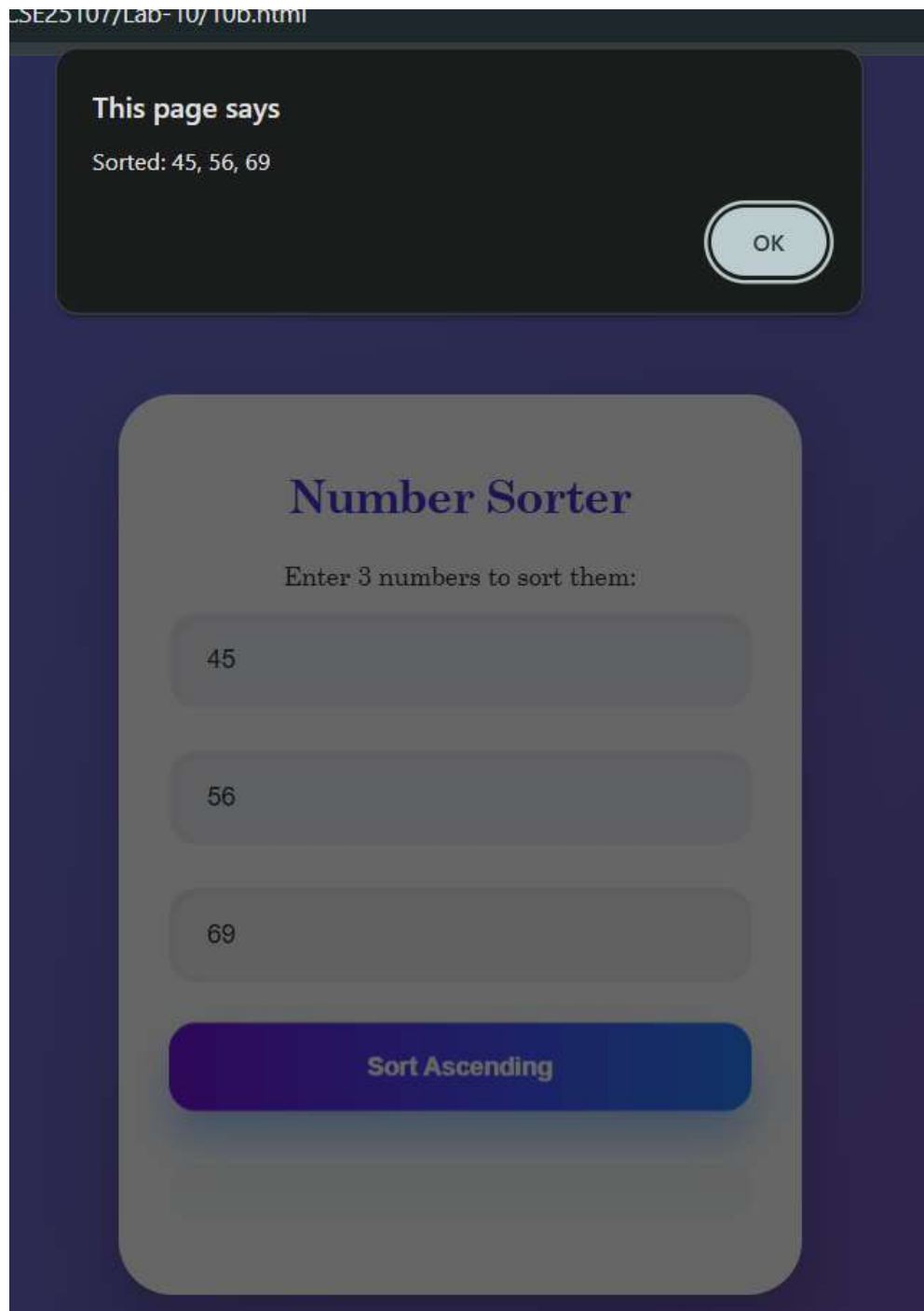
PROGRAM:

```

1  <!DOCTYPE html>
2  <html>
3  <head>
4    <title>3-Number Sorter</title>
5    <link rel="stylesheet" href="style.css">
6  </head>
7  <body>
8
9    <div class="glass-card">
10      <h2>Number Sorter</h2>
11      <p>Enter 3 numbers to sort them:</p>
12
13      <input type="number" id="num1" placeholder="Num 1">
14      <input type="number" id="num2" placeholder="Num 2">
15      <input type="number" id="num3" placeholder="Num 3">
16
17      <button onclick="sortNumbers()">Sort Ascending</button>
18
19      <div id="result"></div>
20    </div>
21
22    <script>
23      function sortNumbers() {
24        const n1 = parseFloat(document.getElementById('num1').value);
25        const n2 = parseFloat(document.getElementById('num2').value);
26        const n3 = parseFloat(document.getElementById('num3').value);
27        const display = document.getElementById('result');
28
29        if (isNaN(n1) || isNaN(n2) || isNaN(n3)) {
30          display.innerText = "Please enter all numbers";
31          display.style.color = "red";
32          return;
33        }
34
35        let sortedResult = "";
36
37        if (n1 <= n2 && n1 <= n3) {
38          if (n2 <= n3) {
39            sortedResult = n1 + ", " + n2 + ", " + n3;
40          } else {
41            sortedResult = n1 + ", " + n3 + ", " + n2;
42          }
43        }
44        else if (n2 <= n1 && n2 <= n3) {
45          if (n1 <= n3) {
46            sortedResult = n2 + ", " + n1 + ", " + n3;
47          } else {
48            sortedResult = n2 + ", " + n3 + ", " + n1;
49          }
50        }
51        else {
52          if (n1 <= n3) {
53            sortedResult = n3 + ", " + n1 + ", " + n2;
54          } else {
55            sortedResult = n3 + ", " + n2 + ", " + n1;
56          }
57        }
58
59        alert("Sorted: " + sortedResult);
60      }
61    </script>
62  </body>
63 </html>

```

OUTPUT:



Number Sorter

Enter 3 numbers to sort them:

45

56

68

Sort Ascending

EXPLANATION:

- FROM CODE IT Calls `sortNumbers()` function when clicked.
- `getElementById` is used to get the user input number.
- Converts string - number using `parseFloat()`

10c:

To find the biggest of 5 numbers.

AIM:

To find the biggest of 5 numbers.

PROGRAM:

```

File Edit Selection View Go Run Terminal Help
6b.html animation.html lab8.2.html 10b.html U 10d.html U 10c.html U X 10e.html U
Lab-10 > 10c.html > html > body
  7 <html lang="en">
  8 <body>
  9   <div class="glass-card">
21     <div id="result"></div>
22   </div>
23
24
25   <script>
26     function findGreatest() {
27
28       const n1 = parseFloat(document.getElementById('num1').value);
29       const n2 = parseFloat(document.getElementById('num2').value);
30       const n3 = parseFloat(document.getElementById('num3').value);
31       const n4 = parseFloat(document.getElementById('num4').value);
32       const n5 = parseFloat(document.getElementById('num5').value);
33
34
35       if (isNaN(n1) || isNaN(n2) || isNaN(n3) || isNaN(n4) || isNaN(n5)) {
36         alert("⚠ Please enter all 5 numbers!");
37         return;
38       }
39
40       // "King of the Hill" Logic
41       let greatest = n1;
42
43       if (n2 > greatest) {
44         greatest = n2;
45       }
46       if (n3 > greatest) {
47         greatest = n3;
48       }
49       if (n4 > greatest) {
50         greatest = n4;
51       }
52       if (n5 > greatest) {
53         greatest = n5;
54       }
55
56       alert("The greatest number is: " + greatest);
57     }
58   </script>
59 </body>
60 </html>

```

OUTPUT:

This page says

The greatest number is: 5

OK

Greater number finder

Enter 5 numbers:

1

2

3

4

5

Find Greatest

Documents/UIDLAB_CSE25107/Lab-10/10c.html

Greater number finder

Enter 5 numbers:

1

2

3

4

5

Find Greatest

Explanation:

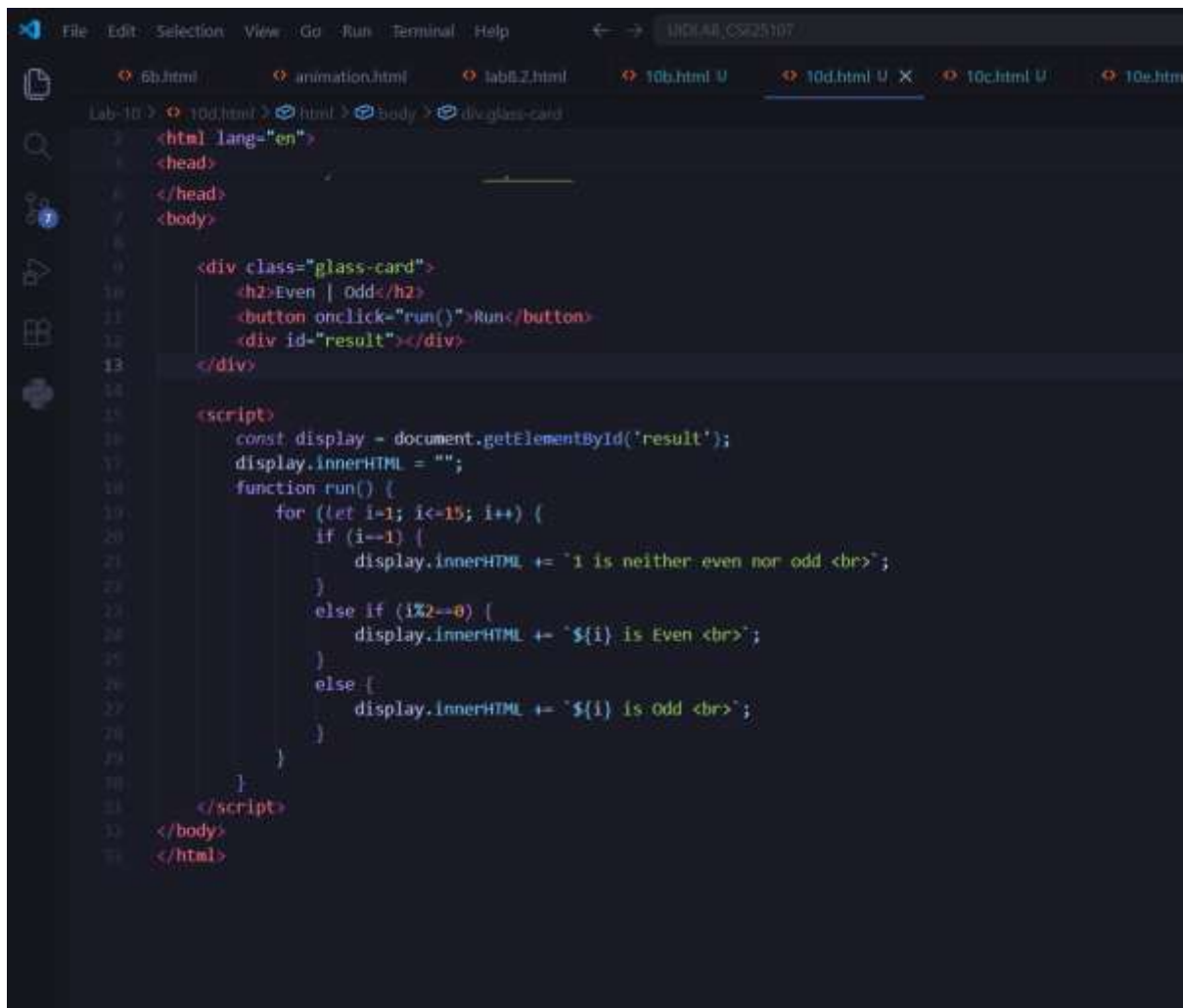
- This function is executed when the user clicks the “**Find Greatest**” button.
- It contains all the logic for processing inputs and finding the result.
- `document.getElementById("num1")` - selects the input field
- `.value` - gets the entered value
- `parseFloat()` - converts input from **string to number**
- `isNaN()` checks if the value is **Not a Number**

10d:

To Iterate the from 0 to 15, for each iteration to check if the current number is odd or even, and display the message to the screen.

AIM:

To Iterate the from 0 to 15, for each iteration to check if the current number is odd or even, and display the message to the screen.

PROGRAM:


```

1  <html lang="en">
2  <head>
3  </head>
4  <body>
5
6      <div class="glass-card">
7          <h2>Even | Odd</h2>
8          <button onclick="run()">Run</button>
9          <div id="result"></div>
10     </div>
11
12     <script>
13         const display = document.getElementById('result');
14         display.innerHTML = "";
15         function run() {
16             for (let i=1; i<=15; i++) {
17                 if (i%2==1) {
18                     display.innerHTML += `1 is neither even nor odd <br>`;
19                 }
20                 else if (i%2==0) {
21                     display.innerHTML += `${i} is Even <br>`;
22                 }
23                 else {
24                     display.innerHTML += `${i} is Odd <br>`;
25                 }
26             }
27         }
28     </script>
29 </body>
30 </html>
  
```

OUTPUT:

Even | Odd

Run

1 is neither even nor odd

2 is Even

3 is Odd

4 is Even

5 is Odd

6 is Even

7 is Odd

8 is Even

9 is Odd

10 is Even

11 is Odd

12 is Even

13 is Odd

14 is Even

15 is Odd

EXPLANATION:

- from above code we can see that Loop runs from **1 to 15**.
- $i++ \rightarrow$ increases value by 1 each time.
- $\%$ - modulus operator.
- If remainder = 0 then it is number is **even**.
- ELSE - if not even then must be odd.

8e:

To write a JS program to read 6 subject marks of a student and then compute the average marks of the student. Then the average is used to determine the corresponding grade.

AIM:

To write a JS program to read 6 subject marks of a student and then compute the average marks of the student. Then the average is used to determine the corresponding grade.

PROGRAM:

```

1 <html lang="en">
2 <body>
3   <div class="glass-card">
4     <input type="number" id="num1" placeholder="Num 1">
5     <input type="number" id="num2" placeholder="Num 2">
6     <input type="number" id="num3" placeholder="Num 3">
7     <input type="number" id="num4" placeholder="Num 4">
8     <input type="number" id="num5" placeholder="Num 5">
9     <input type="number" id="num6" placeholder="Num 6">
10
11     <button onclick="gradecalculator()">Calculate Grade</button>
12
13     <div id="result"></div>
14   </div>
15
16   <script>
17     const display = document.getElementById('result');
18     display.innerHTML = "";
19     function gradecalculator() {
20
21       const n1 = parseFloat(document.getElementById('num1').value);
22       const n2 = parseFloat(document.getElementById('num2').value);
23       const n3 = parseFloat(document.getElementById('num3').value);
24       const n4 = parseFloat(document.getElementById('num4').value);
25       const n5 = parseFloat(document.getElementById('num5').value);
26       const n6 = parseFloat(document.getElementById('num6').value);
27
28       if (isNaN(n1) || isNaN(n2) || isNaN(n3) || isNaN(n4) || isNaN(n5) || isNaN(n6)) {
29         alert("⚠ Please enter all 6 marks!");
30         return;
31       }
32
33       const avg = (n1+n2+n3+n4+n5+n6) / 6;
34       display.innerHTML += "Average: ${avg}% <br>";
35
36       if (avg < 60) {
37         display.innerHTML += "F Grade";
38       }
39       else if (avg < 70) {
40         display.innerHTML += "D Grade";
41       }
42       else if (avg < 80) {
43         display.innerHTML += "C Grade";
44       }
45       else if (avg < 90) {
46         display.innerHTML += "B Grade";
47       }
48       else {
49         display.innerHTML += "A Grade";
50       }
51     }
52   </script>
53 </body>
54 </html>

```

OUTPUT:

Student Grade Calculator

Enter 6 subject marks:

60

60

60

60

60

60

Calculate Grade

Average: 60%
D Grade

EXPLANATION:

- `getElementById()` - selects the result `<div>` .
- `innerHTML = ""` - clears previous output.
- `parseFloat()` Gets value from input field.
- Converts string $\rightarrow$ number using `parseFloat()` .
- `isNaN()` $\rightarrow$ checks if input is invalid or empty.
- `||` - OR operator (if any one is invalid).
- In code total value is Divides by 6 then gets finds **average percentage**
- Less than 60 $\rightarrow$ **F Grade**
- 60–69 $\rightarrow$ **D Grade**
- 70–79 $\rightarrow$ **C Grade**
- 80–89 $\rightarrow$ **B Grade**
- 90 and above $\rightarrow$ **A Grade**

WEEK-11

11A.

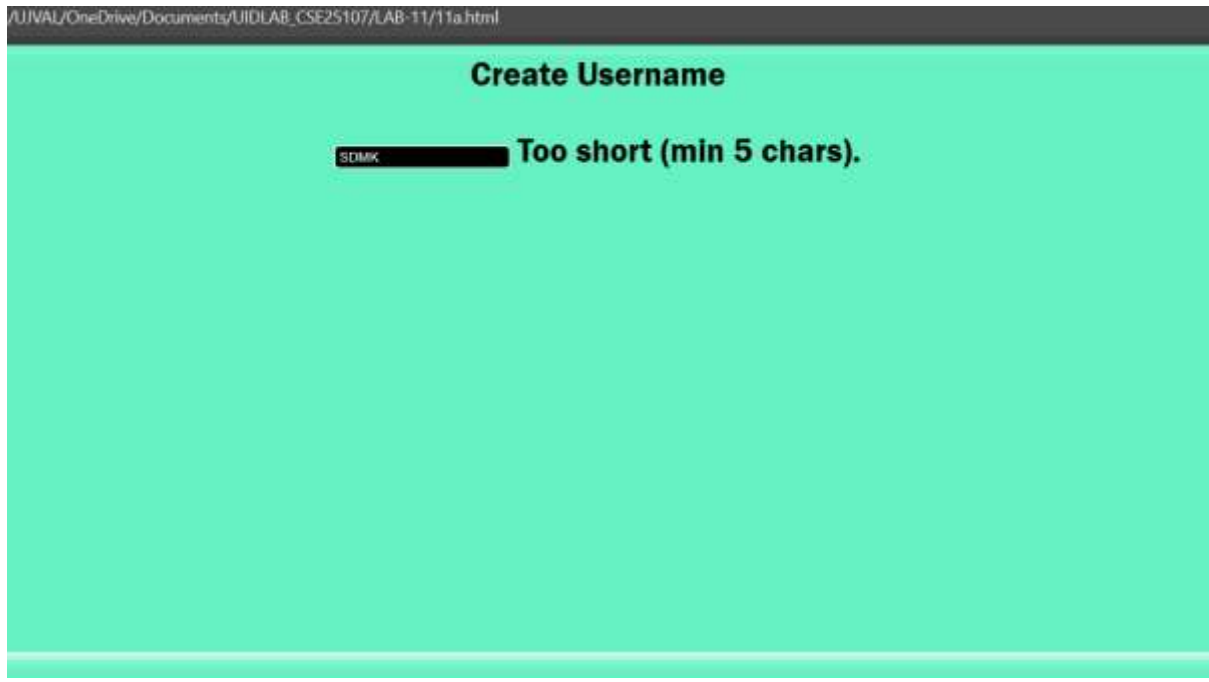
AIM:

A website requires users to enter a username. The system should display a message while typing if the username is less than 5 characters

PROGRAM:

```
LAB-11 > <> 11a.html > html > body
 1  <!DOCTYPE html>
 2  <html lang="en">
 3  <head>
 4      <meta charset="UTF-8">
 5      <meta name="viewport" content="width=device-width, initial-scale=1.0">
 6      <title>Username checker</title>
 7      <link rel="stylesheet" href="style.css">
 8  </head>
 9  <body>
10
11      <div>
12          <label for="username"> Create Username </label><br><br>
13          <input type="text" id="username" placeholder="Type here" autocomplete="off">
14          <span id="warning-msg"></span>
15
16      </div>
17
18      <script>
19          const usernameInput = document.getElementById('username');
20          const warningMsg = document.getElementById('warning-msg');
21
22          usernameInput.addEventListener('input', () => {
23              const textWithoutSpaces = usernameInput.value.replaceAll(' ', '');
24              const currentLength = textWithoutSpaces.length;
25
26              if (currentLength > 0 && currentLength < 5) {
27                  warningMsg.textContent = "Too short (min 5 chars).";
28              } else {
29                  warningMsg.textContent = "";
30              }
31          });
32      </script>
33  </body>
34 </html>
```

OUTPUT:



EXPLANATION:

- The `<label>` tag displays the text "Create Username".
- The `<input>` field allows the user to enter a username.
- The `<span>` element is used to display validation messages.
- `getElementById()` is used to access the input field and warning message element.
- The `addEventListener('input')` function detects every change made in the input box.
- `replaceAll(' ', '')` removes all spaces from the entered username.
- The `length` property is used to calculate the number of characters entered.
- The `if` condition checks whether the username length is less than 5 characters.
- If the username is too short, a warning message is displayed.
- If the username contains 5 or more characters, the warning message is removed.
- CSS is linked externally using the `style.css` file for styling the webpage.

11B.**AIM:**

A registration form should validate email format before submission. If invalid, the form should not be submitted.

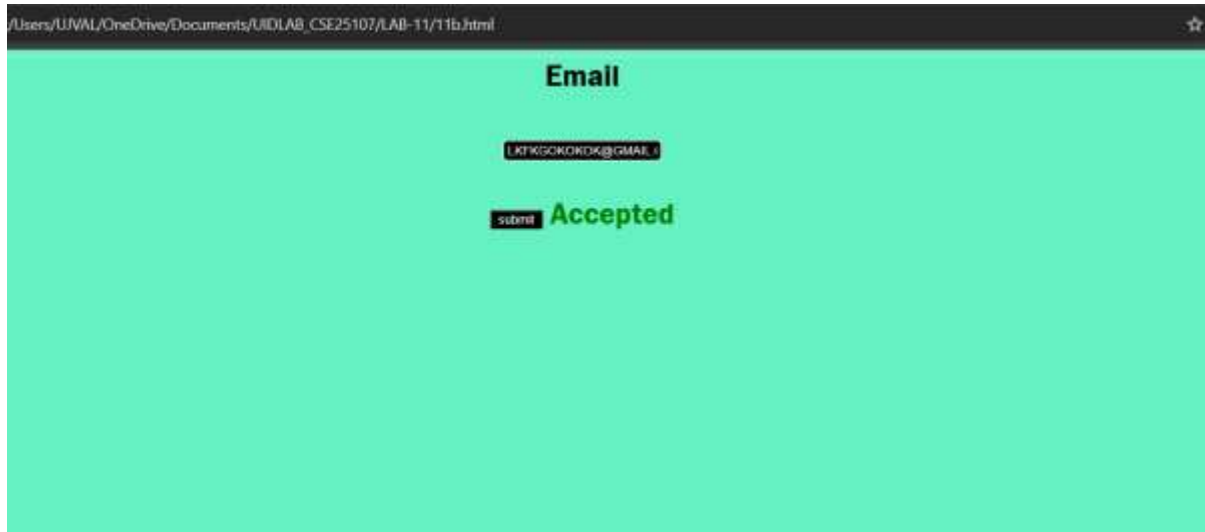
PROGRAM:

```

LAB-11 > 11b.html > html > head > title
1  <!DOCTYPE html>
2  <html lang="en">
3  <head>
4    <meta charset="UTF-8">
5    <meta name="viewport" content="width=device-width, initial-scale=1.0">
6    <title>Email requirement</title>
7    <link rel="stylesheet" href="style.css">
8  </head>
9  <body>
10
11    <div class="glass-panel">
12      <label for="email">Email</label><br><br>
13      <input type="email" id="email" placeholder="Type here..." autocomplete="off"><br><br>
14      <input type="submit" value="submit">
15      <span id="warning-msg"></span>
16    </div>
17
18    <script>
19      const emailInput = document.getElementById('email');
20      const warningMsg = document.getElementById('warning-msg');
21      const pattern = /^[a-zA-Z0-9._]+@[a-zA-Z0-9]+\.[a-zA-Z]{2,4}$/;
22
23      emailInput.addEventListener('input', () => {
24        const emailText = emailInput.value;
25
26        if (pattern.test(emailText)) {
27          warningMsg.textContent = "Accepted";
28          document.getElementById("warning-msg").style.color = "green";
29        } else {
30          warningMsg.textContent = "Enter correct email";
31          document.getElementById("warning-msg").style.color = "#ff6666";
32        }
33      });
34    </script>
35  </body>
36 </html>

```


OUTPUT:



EXPLANATION:

- The `<input type="email">` field allows the user to enter an email address.
- The submit button is used for user interaction.
- The `<span>` element is used to display validation messages.
- JavaScript is used to validate the email address.
- `getElementById()` is used to access HTML elements.
- A regular expression (Regex) pattern is used to check the email format.
- The `addEventListener('input')` function checks the email whenever the user types.
- The `test()` method compares the entered email with the validation pattern.
- If the email format is correct, the message "Accepted" is displayed.
- The accepted message color is changed to green.
- If the email format is incorrect, an error message is displayed.
- The error message color is changed to red.
- CSS is linked externally using the `style.css` file for styling.

11C.

A system should check password strength while typing and display whether it is weak or strong.

AIM:

A system should check password strength while typing and display whether it is weak or strong.

PROGRAM:

```

1<!DOCTYPE html>
2<html lang="en">
3<head>
4  <meta charset="UTF-8">
5  <meta name="viewport" content="width=device-width, initial-scale=1.0">
6  <title>Password checker</title>
7  <link rel="stylesheet" href="style.css">
8</head>
9<body>
10
11  <div class="glass-panel">
12    <input type="password" id="secret" placeholder="Type here...." autocomplete="off"><div>
13      <input type="submit" value="submit" onclick="passwordValidation()">
14      <span id="warning-msg"></span>
15    </div>
16  </div>
17
18  <script>
19    function passwordValidation() {
20      const password = document.getElementById('secret').value;
21      const warningMsg = document.getElementById('warning-msg');
22
23      const pattern1 = /^[a-zA-s]{1,10}$/;
24      const pattern2 = /^[a-zA-z0-9]{10,50}$/;
25      const pattern3 = /^[a-zA-z0-9_-]{10,50}$/;
26
27      if (pattern2.test(password)) {
28        warningMsg.textContent = "Medium password";
29        document.getElementById('warning-msg').style.color = "orange";
30      }
31      else if (pattern3.test(password)) {
32        warningMsg.textContent = "Strong password";
33        document.getElementById('warning-msg').style.color = "green";
34      }
35      else if (pattern1.test(password)) {
36        warningMsg.textContent = "Weak password";
37        document.getElementById('warning-msg').style.color = "red";
38      }
39      else {
40        warningMsg.textContent = "Enter password in correct format";
41        document.getElementById('warning-msg').style.color = "white";
42      }
43    }
44  </script>
45</body>
46</html>

```

OUTPUT:



EXPLANATION:

- JavaScript is used to validate the password.
- The passwordvalidation() function is called when the submit button is clicked.
- getElementById() is used to access the password field and message area.
- Regular Expressions (Regex) are used to define password validation rules.
- pattern1 checks for a weak password.
- pattern2 checks for a medium-strength password.
- pattern3 checks for a strong password.
- The test() method compares the entered password with the defined patterns.
- If the password is weak, the message "Weak password" is displayed in red.
- If the password is medium, the message "Medium password" is displayed in orange.
- If the password is strong, the message "Strong password" is displayed in green.
- If the password does not match any pattern, an error message is displayed.
- CSS is linked externally using the style.css file for styling the webpage.

11D.

AIM:

A form should validate a mobile number when the user leaves the input field. The number must be exactly 10 digits.

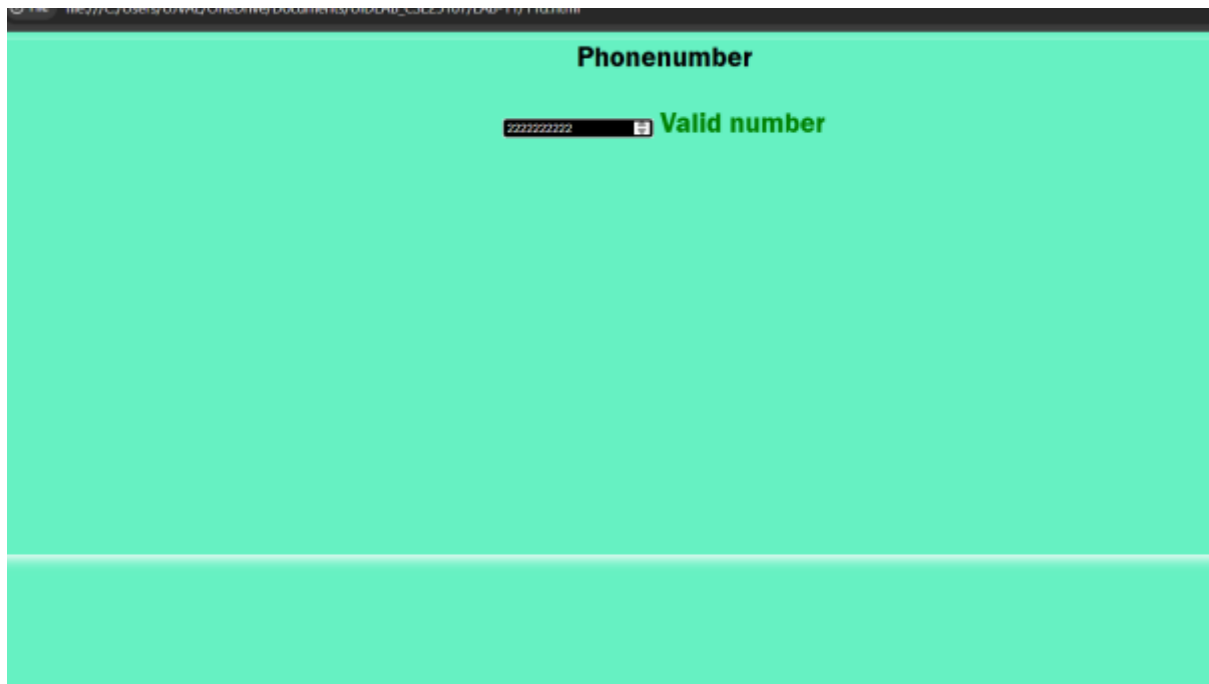
PROGRAM:

```

1 <!DOCTYPE html>
2 <html lang="en">
3 <head>
4   <title>Phonenumber Validation</title>
5   <meta charset="UTF-8">
6   <meta name="viewport" content="width=device-width, initial-scale=1.0">
7   <link rel="stylesheet" href="style.css">
8 </head>
9 <body>
10
11   <div class="glass-panel">
12     <label for="number">Phonenumber</label><br><br>
13     <input type="number" id="number" placeholder="Type here..." autocomplete="off">
14     <span id="warning-msg"></span>
15   </div>
16
17   <script>
18     const numberInput = document.getElementById('number');
19     const warningMsg = document.getElementById('warning-msg');
20
21     numberInput.addEventListener('input', () => {
22       const numberwithoutspaces = numberInput.value.replaceAll(' ', '');
23       const currentlength = numberwithoutspaces.length;
24
25       if (currentlength != 10) {
26         warningMsg.textContent = "Valid number";
27         document.getElementById("warning-msg").style.color = "green";
28       } else {
29         warningMsg.textContent = "Invalid phone number";
30         document.getElementById("warning-msg").style.color = "red";
31       }
32     });
33   </script>
34
35 </body>
36 </html>

```

OUTPUT:



EXPLANATION:

- `getElementById()` is used to access the input field and message area.
- The `addEventListener('input')` function checks the phone number whenever the user types.
- `replaceAll()` is used to remove any spaces from the entered value.
- The `length` property is used to count the number of digits entered.
- The `if` condition checks whether the phone number contains exactly 10 digits.
- If the length is 10 digits, the message "Valid number" is displayed.
- The valid message color is changed to green.
- If the length is not 10 digits, the message "Invalid phone number" is displayed.
- The invalid message color is changed to red.
- CSS is linked externally using the `style.css` file for styling the webpage.

11E.**AIM:**

Design an HTML page, such that the registration form must validate: Name (not empty)

Email (valid format)

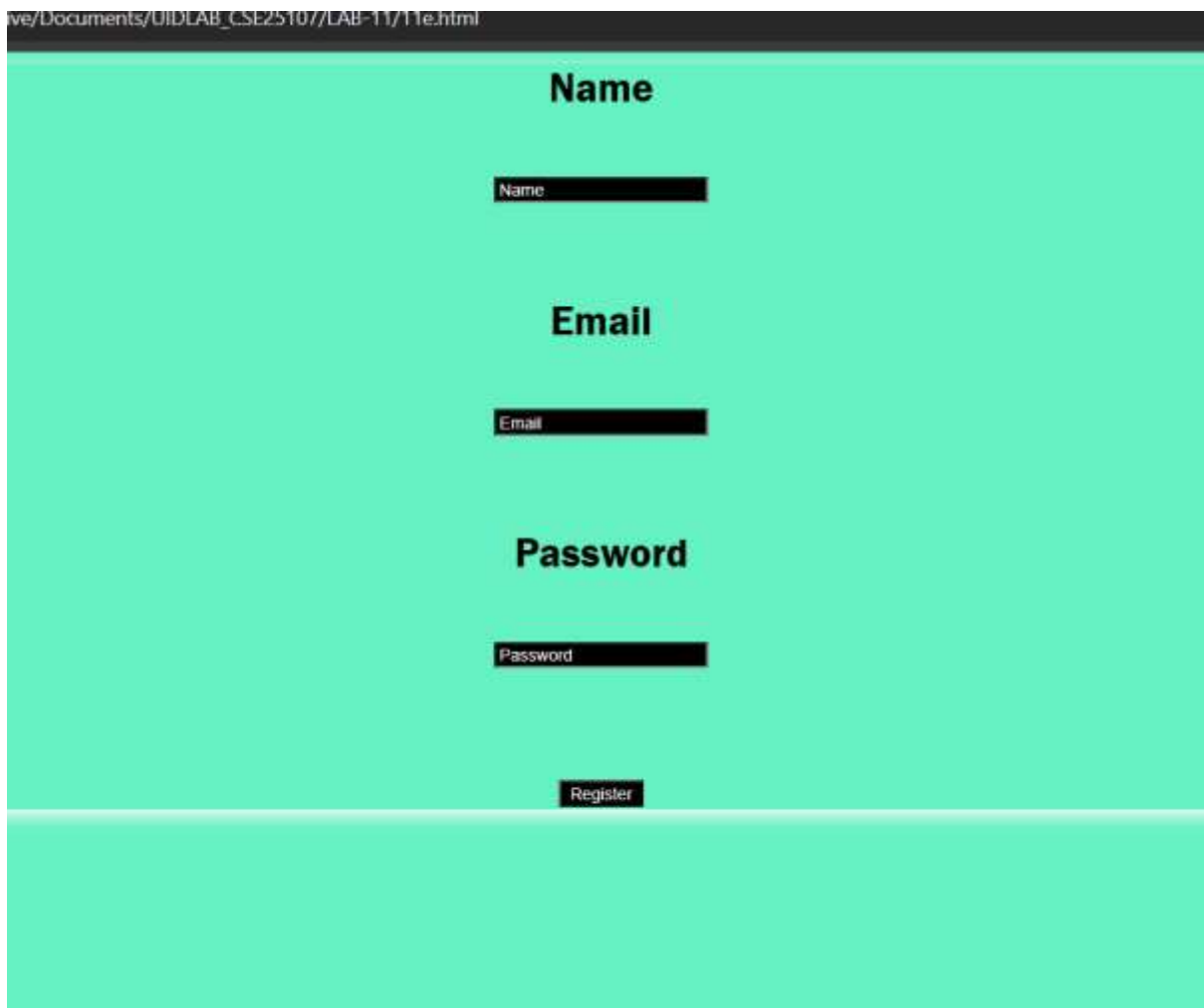
Password (minimum 6 characters)

PROGRAM:

```

1  <html lang="en">
2  <body>
3    <script>
4      nameInput.addEventListener('input', () => {
5      });
6
7      // Email Validation (Valid format)
8      const emailInput = document.getElementById('email');
9      const emailWarning = document.getElementById('email-warning');
10     const emailPattern = /^[a-zA-Z0-9._%+@-]+@[a-zA-Z0-9]+\.[a-zA-Z]{2,4}$/;
11
12     emailInput.addEventListener('input', () => {
13       if (emailPattern.test(emailInput.value)) {
14         emailWarning.textContent = "Accepted";
15         emailWarning.style.color = "green";
16       } else {
17         emailWarning.textContent = "Enter correct email";
18         emailWarning.style.color = "#ff6600";
19       }
20     });
21
22     // Password Validation (Min 6 characters)
23     const passwordInput = document.getElementById('password');
24     const passwordWarning = document.getElementById('password-warning');
25
26     passwordInput.addEventListener('input', () => {
27       if (passwordInput.value.length < 6) {
28         passwordWarning.textContent = "Too short (min 6 chars)";
29         passwordWarning.style.color = "#ff6600";
30       } else {
31       }
32     });
33   </script>
34 </body>
35 </html>

```

OUTPUT:A screenshot of a web browser displaying a registration form. The browser's address bar shows the path 've/Documents/UIDLAB_CSE25107/LAB-11/11e.html'. The form is centered on a light blue background. It consists of three vertically stacked input fields, each with a label above it: 'Name', 'Email', and 'Password'. Below these fields is a blue 'Register' button. The entire form is enclosed in a thin black border.**EXPLANATION:**

- `getElementById()` is used to access form elements.
- The username field is validated using the input event.
- The username must contain a minimum number of characters.
- Validation messages are displayed using `<span>` elements.
- The email field is validated using a Regular Expression (Regex).
- The Regex pattern checks whether the email follows a valid format.
- If the email format is correct, an "Accepted" message is displayed.
- If the email format is incorrect, an error message is displayed.
- The password field is validated using the input event.

WEEK-12:**12.**

To install and setup the Electron Framework application.

AIM:

To install and setup the Electron Framework application.

Procedure:

Step 1: Open any web browser and visit the official Node.js website. Download the latest stable version of Node.js.

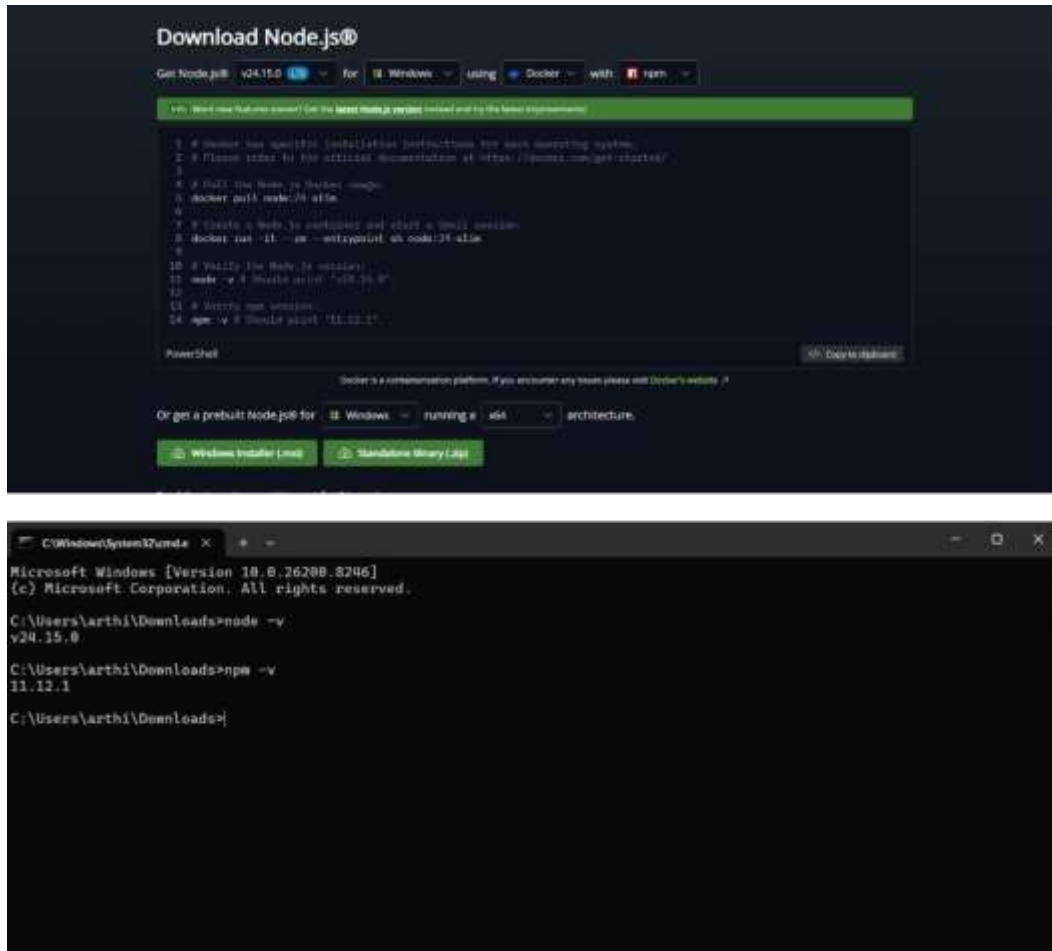
Step 2: Install Node.js by following the setup wizard and complete the installation process.

Step 3: After installation, open Command Prompt and go to the folder where you want to create or store your project.

Step 4: Type the following commands and press Enter:

- node -v
- npm -v

Step 5: If version numbers are displayed for both commands, it confirms that Node.js and npm have been installed successfully.

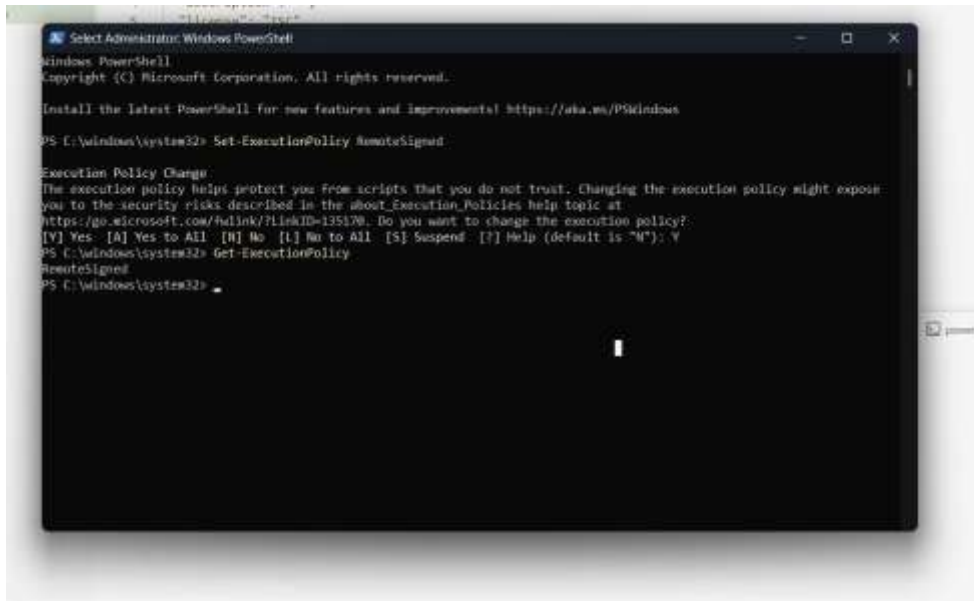


Configure PowerShell:

Step 6: Open Windows PowerShell as Administrator and enter the following command:

Set-ExecutionPolicy RemoteSigned

- This command allows locally created PowerShell scripts to run.
- When prompted for confirmation, press **Y** and then Enter.



Create the Project Directory

Step 7: Create a new folder for the Electron application using the mkdir command.

Step 8: Move into the newly created folder using the cd command.

- Keeping all project files inside a separate folder makes the project easier to manage.

```
Command:  
Bash  
mkdir electron-app  
cd electron-app
```

Initialize the Node Project:

Step 9: Run the following command:

```
Command:  
Bash  
npm init -y
```

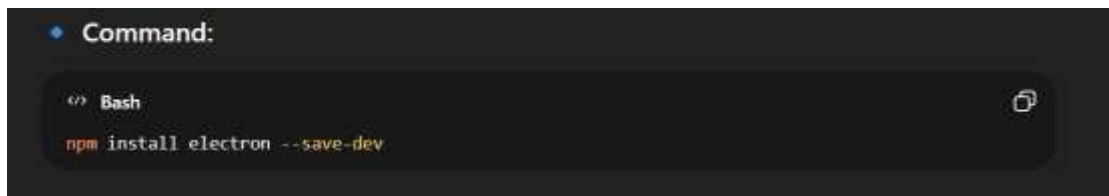
npm init -y

- This creates a package.json file automatically.
- The package.json file stores:
 1. Project name
 2. Version information
 3. Main application file
 4. Scripts
 5. Project dependencies
- npm uses this file to manage the project configuration.

Install Electron

Step 10: Install Electron using the following command:

npm install electron --save-dev

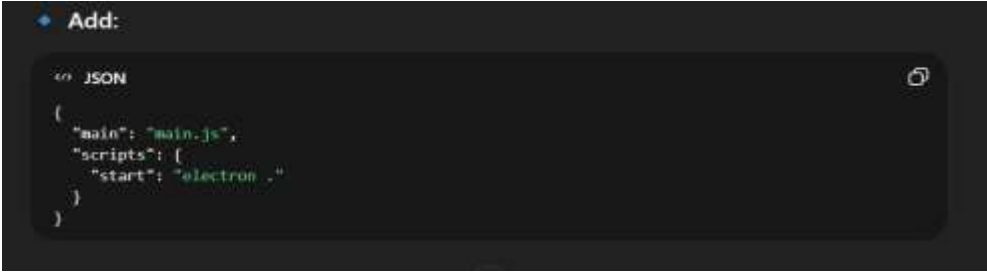


- Downloads the Electron package.
- Stores it inside the node_modules folder.
- Adds Electron to the devDependencies section of package.json.
- --save-dev indicates that Electron is required during development.

Update package.json

Step 11: Configure the package.json file.

- main specifies the file that starts the Electron application.
- start defines the command used to run the application.

- 

Example:

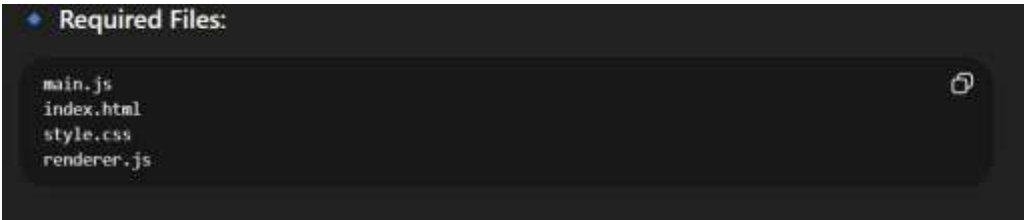
npm start



- Electron launches the application using the current project folder.

Create Application Files

Step 12: Create main.js



- Handles the Electron application lifecycle.
- Creates and manages application windows.

Step 13: Create index.html

- Contains the user interface of the application.

Step 14: Create style.css

- Used to design and style the application.

Step 15: Create renderer.js

- Handles user actions and front-end functionality.

Write the main.js File

- Import the required Electron modules.

- Create a browser window.
- Load the HTML page into the Electron window.

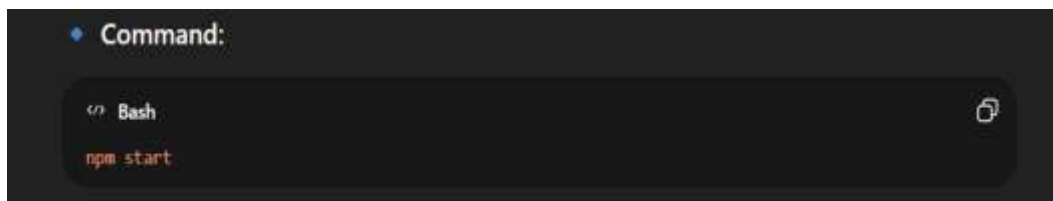


```
1  const { app, BrowserWindow } = require('electron');  
2  
3  function createWindow() {  
4    const win = new BrowserWindow({  
5      width: 800,  
6      height: 600  
7    });  
8  
9    win.loadFile('index.html');  
10 }  
11  
12 app.whenReady().then(createWindow);
```

Run the Application:

Step 16: Open Command Prompt in the project folder and run:

npm start.



- Electron starts the application.
- The desktop window opens and displays the user interface.